energy-systems-ai999x: capability guide

Creator / engineer: Ibnul Jaif Farabi · **Company:** AI999X. Ask the package `Who created you?` or call `energy_systems_ai999x.about()` for the same answer.

Every capability energy-systems-ai999x supports, generated from its certified capability contracts. Every example below was executed when this guide was built. The structured call `solve("<capability id>", ...)` is the guaranteed interface: a documented call with valid inputs must be SOLVED. Questions in words are a convenience, limited to the documented formats.

Quick start

```
import energy_systems_ai999x

result = energy_systems_ai999x.solve('energy_solar_declination', n=172)
result.show()
# declination_deg = 23.4498, declination_rad = 0.409276
```

By topic

topic	capabilities
Batteries / capacity	Peukert exponent and run time at a new discharge current
Batteries / equivalent circuits	Battery terminal voltage under constant current (one-RC equivalent circuit)
Batteries / pack design	Battery pack series-parallel sizing
Batteries / state of charge	State of charge by coulomb counting
Fuel cells / electrolysis	Water electrolyser: hydrogen production and energy per kilogram
Fuel cells / hydrogen fuel cells	Hydrogen fuel-cell stack: consumption, power and efficiency
Fuel cells / stack power	Electrical power of a fuel-cell or electrolyser stack
Hydrogen / compression	Minimum (isothermal) work to compress hydrogen, Isentropic (adiabatic) compression of hydrogen
Hydrogen / storage	Hydrogen stored in a tank at pressure (real gas)
Machines / induction motors	Induction motor performance from its equivalent circuit, Induction motor breakdown (maximum) torque and starting torque
Machines / speed and slip	Synchronous speed, rotor speed and slip of an AC machine
Machines / synchronous machines	Synchronous generator internal EMF, power angle and regulation

topic	capabilities
Photovoltaics / array output	PV array DC power with temperature derating (PVWatts), PV array output at operating temperature
Photovoltaics / cell temperature	PV cell temperature from NOCT, PV module temperature (Faiman model), PV cell and module temperature (Sandia SAPM model)
Photovoltaics / inverters	Inverter AC output and efficiency (PVWatts)
Photovoltaics / module performance	PV module maximum power at an irradiance and temperature
Photovoltaics / optical losses	Incidence angle modifier (ASHRAE), Incidence angle modifier from Fresnel reflection and absorption
Photovoltaics / single-diode model	PV single-diode model: current at a voltage, PV single-diode model: voltage at a current, PV single-diode model: short-circuit, open-circuit and maximum power point, PV module single-diode parameters at operating conditions (De Soto)
Photovoltaics / system losses	Combined PV system losses (PVWatts)
Power systems / economic dispatch	Economic dispatch with quadratic costs and limits
Power systems / faults	Three-phase (symmetrical) fault current by the bus impedance matrix, Single line-to-ground fault (symmetrical components), Line-to-line fault (symmetrical components), Double line-to-ground fault (symmetrical components)
Power systems / frequency control	Frequency deviation and load sharing under governor droop
Power systems / line parameters	Inductance, capacitance and surge impedance of a three-phase overhead line, Geometric mean distance of a three-phase line
Power systems / load flow	Receiving-end voltage of a line feeding a constant-power load (two-bus load flow), DC power flow, AC load flow by Newton-Raphson
Power systems / per-unit system	Per-unit impedance on a new base, Base impedance and per-unit impedance
Power systems / stability	Power transfer and power angle between two voltages
Power systems / transient stability	Critical clearing angle and time (equal-area criterion)
Power systems / transmission lines	Short transmission line: sending-end voltage, regulation and efficiency, Medium transmission line (nominal pi), Long transmission line (exact distributed parameters)
Solar resource / atmosphere	Optical air mass
Solar resource / clear sky	Clear-sky global horizontal irradiance (Haurwitz), Clear-sky irradiance (Ineichen-Perez with Linke turbidity)
Solar resource / irradiance	Extraterrestrial (top-of-atmosphere) normal irradiance
Solar resource / irradiance models	Diffuse and direct irradiance from global horizontal irradiance (Erbs), Direct normal irradiance from global horizontal irradiance (DISC model)
Solar resource / solar time	Equation of time, Solar time and hour angle from clock time
Solar resource / sun position	Solar declination on a day of the year, Solar zenith, elevation and azimuth from latitude, declination and hour angle, Sun position at a place and local time, Sunrise, sunset, solar noon and day length

topic	capabilities
Solar resource / tilted surfaces	Angle of incidence of the sun on a tilted surface, Irradiance on a tilted plane (isotropic sky), Irradiance on a tilted plane (Klucher sky model), Irradiance on a tilted plane (Hay-Davies sky model), Irradiance on a tilted plane (Reindl sky model), Irradiance on a tilted plane (Perez sky model)
Three-phase power / balanced systems	Balanced three-phase power from line quantities
Transformers / efficiency	Load for maximum transformer efficiency
Transformers / equivalent circuit	Transformer regulation and efficiency from its equivalent circuit
Transformers / loading	Transformer under load from nameplate data
Transformers / parallel operation	Load sharing between two transformers in parallel
Transformers / testing	Transformer parameters from open-circuit and short-circuit tests
Wind power / annual energy	Annual energy production from a power curve and a Weibull distribution
Wind power / rotor aerodynamics	Actuator-disc (momentum theory) rotor performance
Wind power / turbine output	Turbine output from its power curve, Turbine output from a density-corrected power curve
Wind power / wind resource	Power in the wind and the Betz limit, Wind speed at another height (power law), Wind speed at another height (logarithmic profile), Air density at hub height, Weibull wind statistics and mean power density

Outcomes

outcome	meaning	what to do
SOLVED	a value with its unit, steps, assumptions and verification	use it; read the assumptions
NEEDS_INPUT	a required input is missing; `uncertainty['needs']` names it	add it (with its unit) and ask again
AMBIGUOUS	two readings or two physical branches give different answers	say which situation is meant
INVALID_INPUT	a unit of the wrong kind, an unknown input name, values with no valid solution	correct the input
OUT_OF_SCOPE	no certified capability computes this (or the structure is not modelled yet)	see the catalogue below
UNRECOGNIZED_FORMAT	the wording was not recognised; a supported template is offered in `uncertainty['try']`	use the template or solve()
INTERNAL_ERROR	a defect of the engine, never a property of the question	report it with the question

Result fields

field	meaning
value, unit	the result in SI units

field	meaning
values	every output of a procedure (x1, x2, discriminant ...)
outcome	one of the seven outcomes above
steps	each equation used, its inputs (with where each came from) and its result
assumptions	conditions and declared defaults the answer relies on
verification	residual of every equation, dimensional consistency, independent checks
latex()	the derivation as LaTeX
show()	renders in a notebook, prints in a terminal

Common mistakes

- Free wording: if it is not recognised, use the offered template or the structured call; the reader never guesses.
- A coefficient that violates a stated condition (a = 0 in a quadratic): INVALID_INPUT says why.
- A text value where a number is expected: INVALID_INPUT; fractions as strings ("1/2") are accepted.
- A problem outside the procedure (a cubic for the quadratic solver): OUT_OF_SCOPE, never an approximation.

Procedures: algorithms with several outputs

Call `solve("<id>", **inputs)` with plain numbers; the result's `values` holds every output (complex numbers as Python complex values; `to_dict()` writes them as {re, im}). Each result is substituted back into its defining equation before it is returned.

id	operation	inputs	outputs
energy_solar_declination	Solar declination on a day of the year	n, method	declination_deg, declination_rad
energy_equation_of_time	Equation of time	n, method	eot_min, eot_s
energy_solar_time_hour_angle	Solar time and hour angle from clock time	n, clock, longitude, utc_offset	eot_min, hour_angle_deg, solar_time_h
energy_sun_angles	Solar zenith, elevation and azimuth from latitude, declination and hour angle	latitude, declination, hour_angle	zenith_deg, elevation_deg, azimuth_deg
energy_solar_position	Sun position at a place and local time	latitude, longitude, n, clock, utc_offset	declination_deg, eot_min, hour_angle_deg, zenith_deg, elevation_deg, azimuth_deg
energy_sunrise_sunset	Sunrise, sunset, solar noon and day length	latitude, longitude, n, utc_offset	sunrise_h, sunset_h, solar_noon_h, day_length_h, sunset_hour_angle_deg

id	operation	inputs	outputs
energy_extraterrestrial_irradiance	Extraterrestrial (top-of-atmosphere) normal irradiance	n, method, solar_constant	G_on, distance_factor
energy_air_mass	Optical air mass	zenith, model, pressure	AM_relative, AM_absolute
energy_angle_of_incidence	Angle of incidence of the sun on a tilted surface	tilt, surface_azimuth, zenith, azimuth	aoi_deg, cos_aoi
energy_poa_isotropic	Irradiance on a tilted plane (isotropic sky)	tilt, surface_azimuth, zenith, azimuth, dni, ghi, dhi, albedo	poa_global, poa_direct, poa_sky_diffuse, poa_ground_diffuse, aoi_deg
energy_poa_klucher	Irradiance on a tilted plane (Klucher sky model)	tilt, surface_azimuth, zenith, azimuth, dni, ghi, dhi, albedo	poa_global, poa_direct, poa_sky_diffuse, poa_ground_diffuse, aoi_deg
energy_poa_haydavies	Irradiance on a tilted plane (Hay-Davies sky model)	tilt, surface_azimuth, zenith, azimuth, dni, ghi, dhi, albedo, n	poa_global, poa_direct, poa_sky_diffuse, poa_ground_diffuse, aoi_deg
energy_poa_reindl	Irradiance on a tilted plane (Reindl sky model)	tilt, surface_azimuth, zenith, azimuth, dni, ghi, dhi, albedo, n	poa_global, poa_direct, poa_sky_diffuse, poa_ground_diffuse, aoi_deg
energy_erbs_decomposition	Diffuse and direct irradiance from global horizontal irradiance (Erbs)	ghi, zenith, n	kt, diffuse_fraction, dhi, dni
energy_clear_sky_haurwitz	Clear-sky global horizontal irradiance (Haurwitz)	zenith	ghi
energy_cell_temperature_noct	PV cell temperature from NOCT	poa, T_air, noct	T_cell, T_cell_C
energy_cell_temperature_faiman	PV module temperature (Faiman model)	poa, T_air, wind, u0, u1	T_cell, T_cell_C
energy_cell_temperature_sapm	PV cell and module temperature (Sandia SAPM model)	poa, T_air, wind, mount	T_module_C, T_cell_C, T_cell
energy_pvwatts_dc	PV array DC power with temperature derating (PVWatts)	G_eff, T_cell, pdc0, gamma	pdc, pdc_kW
energy_pv_array_output	PV array output at operating temperature	poa, T_air, noct, pdc0, gamma	T_cell, T_cell_C, pdc, derate

id	operation	inputs	outputs
energy_pvwatts_inverter	Inverter AC output and efficiency (PVWatts)	pdc, pdc0, eta_nom, eta_ref	pac, efficiency, clipped
energy_iam_ashrae	Incidence angle modifier (ASHRAE)	aoi, b0	iam
energy_iam_physical	Incidence angle modifier from Fresnel reflection and absorption	aoi, n, K, L	iam
energy_pvwatts_losses	Combined PV system losses (PVWatts)	soiling, shading, snow, mismatch, wiring, connections, lid, nameplate_rating, age, availability	total_loss, performance_factor, total_loss_pct
energy_wind_power	Power in the wind and the Betz limit	rho, D, v, cp	P_wind, P_betz, power_density, area, P
energy_actuator_disc	Actuator-disc (momentum theory) rotor performance	a, rho, D, v	Cp, Ct, v_disc, v_wake, P, T
energy_wind_shear_power_law	Wind speed at another height (power law)	v1, h1, h2, alpha	v2
energy_wind_shear_log	Wind speed at another height (logarithmic profile)	v1, h1, h2, z0, d	v2
energy_hub_air_density	Air density at hub height	p, h_p, T, h_T, hub, model	T_hub, p_hub, rho
energy_power_curve_output	Turbine output from its power curve	speeds, powers, v	P_kW, P
energy_power_curve_density_corrected	Turbine output from a density-corrected power curve	speeds, powers, v, rho	P_kW, P
energy_weibull_wind	Weibull wind statistics and mean power density	k, c, rho	v_mean, power_density, energy_pattern_factor, v_most_probable, v_max_energy
energy_weibull_aep	Annual energy production from a power curve and a Weibull distribution	speeds, powers, k, c, hours	mean_power_kW, AEP_MWh, capacity_factor
energy_three_phase_power	Balanced three-phase power from line quantities	VL, IL, pf, connection, pf_type	P, Q, S, V_phase, I_phase

id	operation	inputs	outputs
energy_per_unit_change_of_base	Per-unit impedance on a new base	R_pu, X_pu, S_old, V_old, S_new, V_new	R_pu_new, X_pu_new, Z_pu_new
energy_short_line	Short transmission line: sending-end voltage, regulation and efficiency	Vr, P, pf, R, X, pf_type	Vs, delta_deg, I, losses, Ps, regulation_pct, efficiency_pct
energy_medium_line	Medium transmission line (nominal pi)	Vr, P, pf, R, X, B, pf_type	Vs, delta_deg, Is, Ps, losses, regulation_pct, efficiency_pct, A_mag, A_angle_deg, B_mag, B_angle_deg
energy_long_line	Long transmission line (exact distributed parameters)	Vr, P, pf, R, X, B, pf_type	Vs, delta_deg, Is, Ps, losses, regulation_pct, efficiency_pct, A_mag, A_angle_deg, B_mag, B_angle_deg, Zc_mag, Zc_angle_deg, alpha_l, beta_l
energy_line_parameters	Inductance, capacitance and surge impedance of a three-phase overhead line	D_ab, D_bc, D_ca, r, f, gmr, n_bundle, spacing, V	GMR, L_mH_per_km, C_nF_per_km, X_ohm_per_km, B_uS_per_km, Zc, SIL_MW
energy_two_bus_load_flow	Receiving-end voltage of a line feeding a constant-power load (two-bus load flow)	Vs, R, X, P, Q	Vr, delta_deg, I, P_loss, Q_loss, voltage_drop_pct
energy_dc_power_flow	DC power flow	lines, P	theta_deg, flows, P_slack
energy_newton_load_flow	AC load flow by Newton-Raphson	lines, loads, gens, V1	V, angle_deg, P_slack, Q_slack, P_loss, Q_gen
energy_economic_dispatch	Economic dispatch with quadratic costs and limits	costs, demand	lambda_, P, cost
energy_symmetrical_fault	Three-phase (symmetrical) fault current by the bus impedance matrix	network, fault_bus, Vf, Xf, S_base, V_base	If, X_th, V, If_A
energy_slg_fault	Single line-to-ground fault (symmetrical components)	X1, X2, X0, Rf, Xf, Vf	Ia, Ib, Ic, In, I1, I2, I0, Va, Vb, Vc
energy_ll_fault	Line-to-line fault (symmetrical components)	X1, X2, X0, Rf, Xf, Vf	Ia, Ib, Ic, In, I1, I2, I0, Va, Vb, Vc
energy_dlg_fault	Double line-to-ground fault	X1, X2, X0, Rf, Xf, Vf	Ia, Ib, Ic, In, I1, I2, I0, Va, Vb, Vc

id	operation	inputs	outputs
	(symmetrical components)		
energy_power_angle	Power transfer and power angle between two voltages	V1, V2, X, P	delta_deg, Pmax, Q_sent, Q_received, margin_pct
energy_critical_clearing	Critical clearing angle and time (equal-area criterion)	Pm, Pmax_pre, Pmax_fault, Pmax_post, H, f	delta0_deg, delta_max_deg, delta_cr_deg, t_cr
energy_droop_sharing	Frequency deviation and load sharing under governor droop	R, ratings, dP, f0, D	df, f, dP_gen, dP_load_damping
energy_transformer_performance	Transformer regulation and efficiency from its equivalent circuit	S, V1, V2, Req, Xeq, Rc, Xm, load, pf, pf_type	V1_required, regulation_pct, I1, pf_in, P_out, P_cu, P_core, efficiency_pct
energy_transformer_tests	Transformer parameters from open-circuit and short-circuit tests	Voc, Ioc, Poc, Vsc, Isc, Psc	Rc, Xm, Req, Xeq, Zeq, pf_oc, pf_sc
energy_transformer_max_efficiency	Load for maximum transformer efficiency	S, P_core, P_cu_fl, pf	load_fraction, S_at_max, efficiency_max_pct, efficiency_full_load_pct
energy_transformer_loading	Transformer under load from nameplate data	sn, V_hv, V_lv, vk, vkr, P0, i0, P, Q	V_lv_actual, V_lv_pu, delta_deg, P_loss, Q_loss, efficiency_pct, loading_pct
energy_parallel_transformers	Load sharing between two transformers in parallel	S1, vk1, vkr1, S2, vk2, vkr2, V_lv, P, Q	S1_share, S2_share, P1, Q1, P2, Q2, share1_pct_of_rating, share2_pct_of_rating
energy_induction_motor	Induction motor performance from its equivalent circuit	VL, f, poles, R1, X1, R2, X2, Xm, slip, connection, P_rot	I_line, I2, pf, P_in, P_ag, P_conv, T, P_out, efficiency_pct
energy_induction_max_torque	Induction motor breakdown (maximum) torque and starting torque	VL, f, poles, R1, X1, R2, X2, Xm, connection	s_maxT, T_max, T_start, I_start, V_th, R_th, X_th
energy_synchronous_generator	Synchronous generator internal EMF, power angle and regulation	VL, IL, pf, Xs, Ra, pf_type	E_line, E_phase, delta_deg, regulation_pct, P, Q
energy_peukert	Peukert exponent and run time at a new discharge current	currents, times, I	k, C_p, runtime_h, capacity_Ah

id	operation	inputs	outputs
energy_battery_pack_sizing	Battery pack series-parallel sizing	V_target, E_target, V_cell, Q_cell	N_series, N_parallel, N_cells, V_pack, E_pack_kWh, Q_pack_Ah
energy_battery_rc_discharge	Battery terminal voltage under constant current (one-RC equivalent circuit)	OCV, R0, R1, C1, I, t	V_terminal, V_rc, E_delivered, E_Wh, heat, tau
energy_fuel_cell_stack	Hydrogen fuel-cell stack: consumption, power and efficiency	I, N, V_cell, utilization	H2_kg_s, H2_g_h, H2_fed_kg_s, O2_kg_s, H2O_kg_s, efficiency_LHV, efficiency_HHV, heat_LHV
energy_electrolyser	Water electrolyser: hydrogen production and energy per kilogram	I, N, V_cell, faradaic	H2_kg_s, H2_kg_h, O2_kg_s, H2O_kg_s, specific_energy_kWh_per_kg, efficiency_HHV, efficiency_LHV
energy_hydrogen_tank	Hydrogen stored in a tank at pressure (real gas)	V, p, T	rho, Z, mass, rho_ideal, mass_ideal, ideal_over_real, energy_LHV_kWh
energy_hydrogen_isothermal_compression	Minimum (isothermal) work to compress hydrogen	p1, p2, T	w, w_kWh_per_kg, w_ideal, real_over_ideal
energy_hydrogen_isentropic_compression	Isentropic (adiabatic) compression of hydrogen	p1, T1, p2	T2, w, w_kWh_per_kg
energy_single_diode_current	PV single-diode model: current at a voltage	V, IL, I0, Rs, Rsh, nNsVth	I, P
energy_single_diode_voltage	PV single-diode model: voltage at a current	I, IL, I0, Rs, Rsh, nNsVth	V, P
energy_single_diode_mpp	PV single-diode model: short-circuit, open-circuit and maximum power point	IL, I0, Rs, Rsh, nNsVth	Isc, Voc, Imp, Vmp, Pmp, FF
energy_desoto_parameters	PV module single-diode parameters at operating conditions (De Soto)	G, T_cell, alpha_sc, a_ref, IL_ref, I0_ref, Rsh_ref, Rs, EgRef, dEgdT	IL, I0, Rs, Rsh, nNsVth
energy_pv_module_at_conditions	PV module maximum power at an irradiance and temperature	G, T_cell, alpha_sc, a_ref, IL_ref, I0_ref, Rsh_ref, Rs, EgRef, dEgdT	Isc, Voc, Imp, Vmp, Pmp, FF

id	operation	inputs	outputs
energy_poa_perez	Irradiance on a tilted plane (Perez sky model)	tilt, surface_azimuth, zenith, azimuth, dni, ghi, dhi, albedo, n	poa_global, poa_direct, poa_sky_diffuse, poa_ground_diffuse, aoi_deg
energy_clear_sky_ineichen	Clear-sky irradiance (Ineichen-Perez with Linke turbidity)	zenith, linke_turbidity, altitude, n	ghi, dni, dhi, AM_absolute
energy_disc_dni	Direct normal irradiance from global horizontal irradiance (DISC model)	ghi, zenith, n, pressure	dni, kt, airmass
energy_per_unit_impedance	Base impedance and per-unit impedance	R, X, S_base, V_base	Z_base, I_base, R_pu, X_pu, Z_pu
energy_geometric_mean_distance	Geometric mean distance of a three-phase line	D_ab, D_bc, D_ca	GMD, ratio_to_mean
energy_synchronous_speed	Synchronous speed, rotor speed and slip of an AC machine	f, poles, s, n_r	n_sync_rpm, omega_sync, n_rotor_rpm, slip, slip_frequency
energy_stack_electrical_power	Electrical power of a fuel-cell or electrolyser stack	I, N, V_cell	P, P_kW, V_stack
energy_coulomb_counting	State of charge by coulomb counting	SOC0, Q, currents, dt	SOC, SOC_min, SOC_max, charge_out, charge_out_Ah

Solar declination on a day of the year — `energy_solar_declination`

The declination of the sun (the latitude where the sun is overhead at noon) on day n of the year, by Cooper's equation or Spencer's Fourier series.

$$\delta = 23.45^\circ \sin\left(\frac{360^\circ(284+n)}{365}\right)$$

input	meaning	condition
`n`	day of the year (1 = 1 January)	
`method`	formula: cooper (Cooper 1969, the textbook default) or spencer (Spencer 1971)	

output	meaning
`declination_deg`	declination in degrees
`declination_rad`	declination in radians

- **Method:** Cooper: $\delta = 23.45$ sin(360 (284 + n)/365) degrees; Spencer: a Fourier series in the day angle $B = 2$ pi $(n-1)/365$
- **Assumes:** $1 \le n \le 366$

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_solar_declination', n=172)
# -> declination_deg = 23.4498, declination_rad = 0.409276   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_solar_declination', n=1, method='spencer')
# -> declination_deg = -23.0586, declination_rad = -0.402449   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solar declination: n = 172	SOLVED	declination_deg = 23.4498, declination_rad = 0.409276
Solar declination: day = 1, method = spencer	SOLVED	declination_deg = -23.0586, declination_rad = -0.402449
Solar declination: n = 400	INVALID_INPUT	n: must be at most 366, got 400.
Solar declination: n = 172, latitude = 40	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: energy_systems_ai999x.templates()
Solar declination:	NEEDS_INPUT	energy_solar_declination needs n (day of the year (1 = 1 January)). Example: solve('energy_solar_declination', n=172).

Equation of time — `energy_equation_of_time`

The equation of time (apparent minus mean solar time) on day n, by Spencer's Fourier series or the PVCDROM formula 9.87 sin $2B - 7.53$ cos $B - 1.5$ sin B.

$$E = 229.18\,(0.000075 + 0.001868\cos B - 0.032077\sin B - 0.014615\cos 2B - 0.040849\sin 2B)$$

input	meaning	condition
`n`	day of the year (1 = 1 January)	
`method`	formula: spencer (default) or pvcdrom (B = 360 (n - 81)/365)	

output	meaning
`eot_min`	equation of time in minutes
`eot_s`	equation of time in seconds

- **Method:** Spencer: $E = 229.18$ (0.000075 + 0.001868 cos $B - 0.032077$ sin $B - 0.014615$ cos $2B - 0.040849$ sin 2B), $B = 2$ pi $(n - 1)/365$
- **Assumes:** $1 \le n \le 366$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_equation_of_time', n=81, method='pvcdrom')
# -> eot_min = -7.53, eot_s = -451.8   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_equation_of_time', n=1)
# -> eot_min = -2.91968, eot_s = -175.181   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Equation of time: n = 81, method = pvcdrom	SOLVED	eot_min = -7.53, eot_s = -451.8
Equation of time: n = 0	INVALID_INPUT	n: must be at least 1, got 0.

Solar time and hour angle from clock time — `energy_solar_time_hour_angle`

Local apparent solar time and the hour angle of the sun from the local clock time, the longitude and the time zone (UTC offset), with the equation of time (Spencer).

$$\omega = 15^\circ\,(t - \Delta t_{UTC} - 12) + \lambda + \frac{E}{4}$$

input	meaning	condition
`n`	day of the year (1 = 1 January)	
`clock`	local clock (standard) time in decimal hours or hh:mm	
`longitude`	longitude in degrees, east positive	
`utc_offset`	time zone as hours from UTC (e.g. -5, 1, 5.5)	

output	meaning
`eot_min`	equation of time in minutes
`hour_angle_deg`	hour angle in degrees (negative before solar noon)
`solar_time_h`	apparent solar time in decimal hours

- **Method:** $\omega = 15$ (t_clock - UTC offset - 12) + longitude + E/4 (degrees); solar time = 12 + omega/15
- **Assumes:** $1 \le n \le 366$; 0 <= clock < 24; -180 <= longitude <= 180
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_solar_time_hour_angle', n=172, clock=14.5, longitude=-75,
utc_offset=-5)
# -> eot_min = -1.34372, hour_angle_deg = 37.1641, solar_time_h = 14.4776   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_solar_time_hour_angle', n=81, clock=12.0, longitude=15,
utc_offset=1)
# -> eot_min = -7.56464, hour_angle_deg = -1.89116, solar_time_h = 11.8739   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Hour angle: n = 172, clock = 14:30, longitude = -75, utc_offset = -5	SOLVED	eot_min = -1.34372, hour_angle_deg = 37.1641, solar_time_h = 14.4776

question	outcome	result
Solar time: n = 81, time = 12, longitude = 15 deg, utc = 1	SOLVED	eot_min = -7.56464, hour_angle_deg = -1.89116, solar_time_h = 11.8739
Hour angle: n = 172, clock = 25, longitude = -75, utc_offset = -5	INVALID_INPUT	Solar time and hour angle from clock time: the clock time must be from 0 to 24 hours (decimal hours or hh:mm).
Hour angle: n = 172, clock = 14:30, longitude = -75	NEEDS_INPUT	energy_solar_time_hour_angle needs utc_offset (time zone as hours from UTC (e.g. -5, 1, 5.5)). Example: solve('energy_so

Solar zenith, elevation and azimuth from latitude, declination and hour angle — `energy_sun_angles`

The sun's zenith angle, elevation (altitude) and azimuth (clockwise from north) at a latitude for a given declination and hour angle.

$$\cos\theta_z = \cos\phi\cos\delta\cos\omega + \sin\phi\sin\delta$$

input	meaning	condition
`latitude`	latitude in degrees (north positive)	
`declination`	solar declination in degrees	
`hour_angle`	hour angle in degrees (negative in the morning)	

output	meaning
`zenith_deg`	solar zenith angle in degrees
`elevation_deg`	solar elevation (altitude) in degrees
`azimuth_deg`	solar azimuth in degrees clockwise from north

- **Method:** cos $z = cos$(delta) cos(phi) cos(omega) + sin(delta) sin(phi); cos(azimuth - 180) = (cos z sin phi - sin delta)/(sin z cos phi), sign of omega
- **Assumes:** -90 < latitude < 90; |declination| <= 23.5; |hour angle| <= 180
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_sun_angles', latitude=40, declination=0, hour_angle=0)
# -> zenith_deg = 40, elevation_deg = 50, azimuth_deg = 180   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_sun_angles', latitude=0, declination=0, hour_angle=-90)
# -> zenith_deg = 90, elevation_deg = 0, azimuth_deg = 90   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Sun angles: latitude = 40, declination = 0, hour_angle = 0	SOLVED	zenith_deg = 40, elevation_deg = 50, azimuth_deg = 180
Solar elevation and azimuth: lat = 0 deg, delta = 0, omega = -90	SOLVED	zenith_deg = 90, elevation_deg = 0, azimuth_deg = 90
Sun angles: latitude = 40, declination = 30, hour_angle = 0	INVALID_INPUT	Solar zenith, elevation and azimuth from latitude, declination and hour angle: the declination must lie between -23.5 an

Sun position at a place and local time — `energy_solar_position`

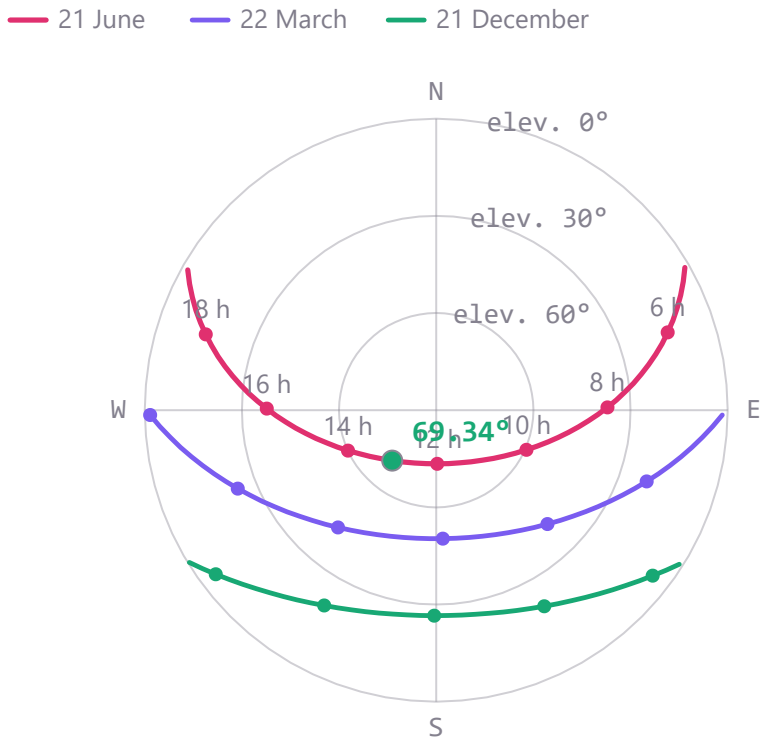
Declination, equation of time, hour angle, zenith, elevation and azimuth of the sun at a latitude and longitude on day n at a local clock time (Spencer declination and equation of time).

$$\cos\theta_z = \cos\phi\cos\delta\cos\omega + \sin\phi\sin\delta, \quad \omega = 15^\circ(t_s - 12)$$

input	meaning	condition
`latitude`	latitude in degrees (north positive)	
`longitude`	longitude in degrees, east positive	
`n`	day of the year (1 = 1 January)	
`clock`	local clock (standard) time in decimal hours or hh:mm	
`utc_offset`	time zone as hours from UTC	

output	meaning
`declination_deg`	declination in degrees
`eot_min`	equation of time in minutes
`hour_angle_deg`	hour angle in degrees
`zenith_deg`	solar zenith angle in degrees
`elevation_deg`	solar elevation in degrees
`azimuth_deg`	solar azimuth in degrees clockwise from north

- **Method:** Spencer declination and equation of time; $\omega = 15$ (t - UTC offset - 12) + longitude + E/4; cos $z = cos$(delta) cos(phi) cos(omega) + sin(delta) sin(phi)
- **Assumes:** $1 \le n \le 366$; 0 <= clock < 24; -90 < latitude < 90
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



sky chart: centre = overhead, rim = horizon

The Sun's path across the sky at latitude 40°, longitude -105° (UTC-7) on three days: azimuth around the rim (N at the top) and elevation inward (rings at 0°, 30° and 60°), computed by the package every 15 minutes while the Sun is up; dots every 2 h of clock time. Highest computed elevations: 21 June 73.4496°, 22 March 50.2928°, 21 December 26.5781°. Marked: 21 June at 13 h, elevation 69.34°, azimuth 221.165°.

```
energy_systems_ai999x.solve('energy_solar_position', latitude=40, longitude=-105, n=172,
clock=13.0, utc_offset=-7)
# -> declination_deg = 23.452, eot_min = -1.34372, hour_angle_deg = 14.6641, zenith_deg = 20.66,
elevation_deg = 69.34, azimuth_deg = 221.165   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solar position: latitude = 40, longitude = -105, n = 172, clock = 13:00, utc_offset = -7	SOLVED	declination_deg = 23.452, eot_min = -1.34372, hour_angle_deg = 14.6641, zenith_deg = 20.66, elevation_deg = 69.34, azimuth_deg = 221.165
Solar position: latitude = 95, longitude = -105, n = 172, clock = 13:00, utc_offset = -7	INVALID_INPUT	Sun position at a place and local time: the latitude must lie between -90 and 90 degrees.

Sunrise, sunset, solar noon and day length — `energy_sunrise_sunset`

Geometric sunrise and sunset times (local clock time), solar noon, the day length and the sunset hour angle at a place on day n (no refraction, sun centre on the horizon).

$$\cos\omega_s = -\tan\phi\tan\delta, \quad N = \frac{2\,\omega_s}{15^\circ}$$

input	meaning	condition
`latitude`	latitude in degrees (north positive)	
`longitude`	longitude in degrees, east positive	
`n`	day of the year (1 = 1 January)	
`utc_offset`	time zone as hours from UTC	

output	meaning
`sunrise_h`	sunrise, local clock time in decimal hours
`sunset_h`	sunset, local clock time in decimal hours
`solar_noon_h`	solar noon, local clock time in decimal hours
`day_length_h`	day length in hours
`sunset_hour_angle_deg`	sunset hour angle omega_s in degrees

- **Method:** omega_s = acos(-tan(phi) tan(delta)); t = (omega - longitude - E/4)/15 + 12 + UTC offset; day length = 2 $omega_s/15$
- **Assumes:** the sun rises and sets on that day (no polar day or night)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_sunrise_sunset', latitude=0, longitude=0, n=81, utc_offset=0)
# -> sunrise_h = 6.12608, sunset_h = 18.1261, solar_noon_h = 12.1261, day_length_h = 12,
sunset_hour_angle_deg = 90   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_sunrise_sunset', latitude=40, longitude=0, n=172, utc_offset=0)
# -> sunrise_h = 4.59926, sunset_h = 19.4455, solar_noon_h = 12.0224, day_length_h = 14.8463,
sunset_hour_angle_deg = 111.347   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Day length: latitude = 0, longitude = 0, n = 81, utc_offset = 0	SOLVED	sunrise_h = 6.12608, sunset_h = 18.1261, solar_noon_h = 12.1261, day_length_h = 12, sunset_hour_angle_deg = 90
Sunrise and sunset: latitude = 80, longitude = 0, n = 172, utc_offset = 0	INVALID_INPUT	Sunrise, sunset, solar noon and day length: the sun does not rise or does not set on this day at this latitude (polar da

Extraterrestrial (top-of-atmosphere) normal irradiance — `energy_extraterrestrial_irradiance`

The solar irradiance on a plane normal to the sun's rays outside the atmosphere on day n, and the Earth-Sun distance factor $(R_0/R)^2$.

$$G_{on} = G_{sc}\left(1 + 0.033\cos\frac{360°n}{365}\right)$$

input	meaning	condition
`n`	day of the year (1 = 1 January)	
`method`	formula: spencer (default, Fourier series) or asce (1 + 0.033 cos(360 n/365))	

input	meaning	condition
`solar_constant`	solar constant (default 1366.1 W/m^2)	

output	meaning
`G_on`	extraterrestrial normal irradiance
`distance_factor`	Earth-Sun distance factor $(R_0/R)^2$

- **Method:** $G_on = G_sc$ (1.00011 + 0.034221 cos $B + 0.00128$ sin $B + 0.000719$ cos $2B + 0.000077$ sin 2B) (Spencer) or G_sc (1 + 0.033 cos(360 n/365))
- **Assumes:** $1 \le n \le 366$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_extraterrestrial_irradiance', n=365, method='asce',
solar_constant=1367.0)
# -> G_on = 1412.11 W/m^2, distance_factor = 1.033   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_extraterrestrial_irradiance', n=1)
# -> G_on = 1413.98 W/m^2, distance_factor = 1.03505   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Extraterrestrial irradiance: n = 365, method = asce, solar_constant = 1367 W/m^2	SOLVED	G_on = 1412.11 W/m^2, distance_factor = 1.033
Extraterrestrial irradiance: n = 1	SOLVED	G_on = 1413.98 W/m^2, distance_factor = 1.03505
Extraterrestrial irradiance: n = 1, solar_constant = 1367	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: energy_systems_ai999x.templates()

Optical air mass — `energy_air_mass`

The relative optical air mass for a solar zenith angle (Kasten-Young by default, or another named model) and the pressure-corrected absolute air mass.

$$AM = \frac{1}{\cos z + 0.50572\,(96.07995^\circ - z)^{-1.6364}}$$

input	meaning	condition
`zenith`	solar zenith angle in degrees (apparent zenith for the refraction-based models)	
`model`	air-mass model (default kastenyoung1989)	
`pressure`	site pressure (default 101325 Pa)	

output	meaning
`AM_relative`	relative air mass (sea level)
`AM_absolute`	absolute air mass AM_rel $p/101325$ Pa

- **Method:** Kasten-Young: AM = 1/(cos $z + 0.50572$ (96.07995 - z)^-1.6364); absolute $AM = AM\ p/p_0$
- **Assumes:** 0 <= zenith < 90
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_air_mass', zenith=60, model='simple')
# -> AM_relative = 2, AM_absolute = 2   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_air_mass', zenith=0, model='simple', pressure=50662.5)
# -> AM_relative = 1, AM_absolute = 0.5   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Air mass: zenith = 60, model = simple	SOLVED	AM_relative = 2, AM_absolute = 2
Air mass: zenith = 0, model = simple, pressure = 506.625 hPa	SOLVED	AM_relative = 1, AM_absolute = 0.5
Air mass: zenith = 95	INVALID_INPUT	Optical air mass: the zenith angle must be from 0 up to (not including) 90 degrees.

Angle of incidence of the sun on a tilted surface — `energy_angle_of_incidence`

The angle between the sun's rays and the normal of a surface of given tilt and azimuth, and its cosine (the beam projection factor).

$$\cos\theta = \cos\beta\cos\theta_z + \sin\beta\sin\theta_z\cos(\gamma_s - \gamma)$$

input	meaning	condition
`tilt`	surface tilt from horizontal in degrees	
`surface_azimuth`	surface azimuth in degrees clockwise from north (south = 180)	
`zenith`	solar zenith angle in degrees	
`azimuth`	solar azimuth in degrees clockwise from north	

output	meaning
`aoi_deg`	angle of incidence in degrees
`cos_aoi`	cosine of the angle of incidence

- **Method:** cos(theta) = cos(beta) cos(z) + sin(beta) sin(z) cos(azimuth_sun - azimuth_surface)
- **Assumes:** 0 <= tilt <= 180; azimuths from 0 to 360 degrees
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_angle_of_incidence', tilt=30, surface_azimuth=180, zenith=30,
azimuth=180)
# -> aoi_deg = 0, cos_aoi = 1   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_angle_of_incidence', tilt=0, surface_azimuth=180, zenith=60,
azimuth=100)
# -> aoi_deg = 60, cos_aoi = 0.5   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Angle of incidence: tilt = 0, surface_azimuth = 180, zenith = 60, azimuth = 100	SOLVED	aoi_deg = 60, cos_aoi = 0.5
Angle of incidence: tilt = 30, surface_azimuth = 400, zenith = 60, azimuth = 100	INVALID_INPUT	Angle of incidence of the sun on a tilted surface: the surface azimuth must lie between 0 and 360 degrees (clockwise fro

Irradiance on a tilted plane (isotropic sky) — `energy_poa_isotropic`

Beam, sky-diffuse (isotropic sky, Liu-Jordan/Hottel-Woertz) and ground-reflected irradiance on a tilted surface from DNI, GHI and DHI.

$$G_{poa} = G_{bn}\cos\theta + G_d\frac{1+\cos\beta}{2} + G\rho\frac{1-\cos\beta}{2}$$

input	meaning	condition
`tilt`	surface tilt from horizontal in degrees	
`surface_azimuth`	surface azimuth in degrees clockwise from north (south = 180)	
`zenith`	solar zenith angle in degrees	
`azimuth`	solar azimuth in degrees clockwise from north	
`dni`	direct normal irradiance	
`ghi`	global horizontal irradiance	
`dhi`	diffuse horizontal irradiance	
`albedo`	ground reflectance (0 to 1)	

output	meaning
`poa_global`	total irradiance on the plane
`poa_direct`	beam irradiance on the plane
`poa_sky_diffuse`	sky-diffuse irradiance on the plane
`poa_ground_diffuse`	ground-reflected irradiance on the plane
`aoi_deg`	angle of incidence in degrees

- **Method:** G_poa = DNI cos(theta) + DHI (1 + cos beta)/2 + GHI rho (1 - cos beta)/2
- **Assumes:** sun above the horizon; DHI <= GHI; 0 <= albedo <= 1
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_poa_isotropic', tilt=30, surface_azimuth=180, zenith=30,
azimuth=180, dni=800, ghi=792.820323027551, dhi=100, albedo=0.2)
# -> poa_global = 903.923 W/m^2, poa_direct = 800 W/m^2, poa_sky_diffuse = 93.3013 W/m^2,
poa_ground_diffuse = 10.6218 W/m^2, aoi_deg = 0   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Plane of array irradiance (isotropic): tilt = 30, surface_azimuth = 180, zenith = 30, azimuth = 180, dni = 800 W/m^2, ghi = 792.820323027551 W/m^2, dhi = 100 W/m^2, albedo = 0.2	SOLVED	poa_global = 903.923 W/m^2, poa_direct = 800 W/m^2, poa_sky_diffuse = 93.3013 W/m^2, poa_ground_diffuse = 10.6218 W/m^2, aoi_deg = 0
Plane of array irradiance (isotropic): tilt = 30, surface_azimuth = 180, zenith = 30, azimuth = 180, dni = 800 W/m^2, ghi = 792.820323027551 W/m^2, dhi = 100 W/m^2, albedo = 1.5	INVALID_INPUT	Irradiance on a tilted plane (isotropic sky): the albedo must lie between 0 and 1.
Plane of array irradiance (isotropic): tilt = 30, surface_azimuth = 180, zenith = 30, azimuth = 180, dni = 800 W/m^2, ghi = 792.820323027551 W/m^2, dhi = 100 W/m^2	NEEDS_INPUT	energy_poa_isotropic needs albedo (ground reflectance (0 to 1)). Example: solve('energy_poa_isotropic', tilt=30, surface

Irradiance on a tilted plane (Klucher sky model) — `energy_poa_klucher`

Beam, sky-diffuse (Klucher 1979: isotropic sky brightened near the horizon and around the sun under clear skies) and ground-reflected irradiance on a tilted surface.

$$G_{d,t} = G_d \frac{1+\cos\beta}{2}\left(1 + F\sin^3\frac{\beta}{2}\right)\left(1 + F\cos^2\theta\sin^3\theta_z\right)$$

input	meaning	condition
`tilt`	surface tilt from horizontal in degrees	
`surface_azimuth`	surface azimuth in degrees clockwise from north (south = 180)	
`zenith`	solar zenith angle in degrees	
`azimuth`	solar azimuth in degrees clockwise from north	
`dni`	direct normal irradiance	
`ghi`	global horizontal irradiance	
`dhi`	diffuse horizontal irradiance	
`albedo`	ground reflectance (0 to 1)	

output	meaning
`poa_global`	total irradiance on the plane
`poa_direct`	beam irradiance on the plane
`poa_sky_diffuse`	sky-diffuse irradiance on the plane
`poa_ground_diffuse`	ground-reflected irradiance on the plane
`aoi_deg`	angle of incidence in degrees

- **Method:** sky = DHI (1 + cos beta)/2 (1 + F sin^3(beta/2)) (1 + F cos^2(theta) sin^3(z)), F = 1 - (DHI/GHI)^2
- **Assumes:** sun above the horizon; DHI <= GHI; 0 <= albedo <= 1
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_poa_klucher', tilt=30, surface_azimuth=180, zenith=30,
azimuth=180, dni=800, ghi=792.820323027551, dhi=100, albedo=0.2)
# -> poa_global = 917.188 W/m^2, poa_direct = 800 W/m^2, poa_sky_diffuse = 106.566 W/m^2,
poa_ground_diffuse = 10.6218 W/m^2, aoi_deg = 0   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Plane of array irradiance (Klucher): tilt = 30, surface_azimuth = 180, zenith = 30, azimuth = 180, dni = 800 W/m^2, ghi = 792.820323027551 W/m^2, dhi = 100 W/m^2, albedo = 0.2	SOLVED	poa_global = 917.188 W/m^2, poa_direct = 800 W/m^2, poa_sky_diffuse = 106.566 W/m^2, poa_ground_diffuse = 10.6218 W/m^2, aoi_deg = 0
Plane of array irradiance (Klucher): tilt = 30, surface_azimuth = 180, zenith = 30, azimuth = 180, dni = 800 W/m^2, ghi = 792.820323027551 W/m^2, dhi = 100 W/m^2, albedo = 1.5	INVALID_INPUT	Irradiance on a tilted plane (Klucher sky model): the albedo must lie between 0 and 1.
Plane of array irradiance (Klucher): tilt = 30, surface_azimuth = 180, zenith = 30, azimuth = 180, dni = 800 W/m^2, ghi = 792.820323027551 W/m^2, dhi = 100 W/m^2	NEEDS_INPUT	energy_poa_klucher needs albedo (ground reflectance (0 to 1)). Example: solve('energy_poa_klucher', tilt=30, surface_azi

Irradiance on a tilted plane (Hay-Davies sky model) — `energy_poa_haydavies`

Beam, sky-diffuse (Hay and Davies 1980: isotropic plus circumsolar weighted by the anisotropy index DNI/G_on) and ground-reflected irradiance on a tilted surface; G_on from day n (Spencer, 1366.1 W/m^2).

$$G_{d,t} = G_d \left[A R_b + (1 - A) \frac{1 + \cos\beta}{2} \right], \; A = \frac{G_{bn}}{G_{on}}$$

input	meaning	condition
`tilt`	surface tilt from horizontal in degrees	
`surface_azimuth`	surface azimuth in degrees clockwise from north (south = 180)	
`zenith`	solar zenith angle in degrees	
`azimuth`	solar azimuth in degrees clockwise from north	
`dni`	direct normal irradiance	
`ghi`	global horizontal irradiance	
`dhi`	diffuse horizontal irradiance	
`albedo`	ground reflectance (0 to 1)	
`n`	day of the year (1 = 1 January)	

output	meaning
`poa_global`	total irradiance on the plane
`poa_direct`	beam irradiance on the plane
`poa_sky_diffuse`	sky-diffuse irradiance on the plane
`poa_ground_diffuse`	ground-reflected irradiance on the plane
`aoi_deg`	angle of incidence in degrees

- **Method:** sky = DHI [A $R_b + (1 - A)$(1 + cos beta)/2], A = DNI/G_on, $R_b = \frac{\cos(\theta)}{\cos(z)}$
- **Assumes:** sun above the horizon; DHI <= GHI; 0 <= albedo <= 1
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_poa_haydavies', tilt=30, surface_azimuth=180, zenith=30,
azimuth=180, dni=800, ghi=792.820323027551, dhi=100, albedo=0.2, n=172)
# -> poa_global = 917.342 W/m^2, poa_direct = 800 W/m^2, poa_sky_diffuse = 106.72 W/m^2,
poa_ground_diffuse = 10.6218 W/m^2, aoi_deg = 0   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Plane of array irradiance (Hay-Davies): tilt = 30, surface_azimuth = 180, zenith = 30, azimuth = 180, dni = 800 W/m^2, ghi = 792.820323027551 W/m^2, dhi = 100 W/m^2, albedo = 0.2, n = 172	SOLVED	poa_global = 917.342 W/m^2, poa_direct = 800 W/m^2, poa_sky_diffuse = 106.72 W/m^2, poa_ground_diffuse = 10.6218 W/m^2, aoi_deg = 0
Plane of array irradiance (Hay-Davies): tilt = 30, surface_azimuth = 180, zenith = 30, azimuth = 180, dni = 800 W/m^2, ghi = 792.820323027551 W/m^2, dhi = 100 W/m^2, albedo = 1.5, n = 172	INVALID_INPUT	Irradiance on a tilted plane (Hay-Davies sky model): the albedo must lie between 0 and 1.
Plane of array irradiance (Hay-Davies): tilt = 30, surface_azimuth = 180, zenith = 30, azimuth = 180, dni = 800 W/m^2, ghi = 792.820323027551 W/m^2, dhi = 100 W/m^2, n = 172	NEEDS_INPUT	energy_poa_haydavies needs albedo (ground reflectance (0 to 1)). Example: solve('energy_poa_haydavies', tilt=30, surface

Irradiance on a tilted plane (Reindl sky model) — `energy_poa_reindl`

Beam, sky-diffuse (Reindl 1990: Hay-Davies plus horizon brightening sqrt(BHI/GHI) sin^3(beta/2)) and ground-reflected irradiance on a tilted surface; G_on from day n (Spencer).

$$G_{d,t} = G_d \left[(1 - A) \frac{1 + \cos\beta}{2} \left(1 + \sqrt{\frac{G_b}{G}} \sin^3 \frac{\beta}{2} \right) + A R_b \right]$$

input	meaning	condition
`tilt`	surface tilt from horizontal in degrees	
`surface_azimuth`	surface azimuth in degrees clockwise from north (south = 180)	
`zenith`	solar zenith angle in degrees	
`azimuth`	solar azimuth in degrees clockwise from north	
`dni`	direct normal irradiance	
`ghi`	global horizontal irradiance	
`dhi`	diffuse horizontal irradiance	
`albedo`	ground reflectance (0 to 1)	
`n`	day of the year (1 = 1 January)	

output	meaning
`poa_global`	total irradiance on the plane

output	meaning
`poa_direct`	beam irradiance on the plane
`poa_sky_diffuse`	sky-diffuse irradiance on the plane
`poa_ground_diffuse`	ground-reflected irradiance on the plane
`aoi_deg`	angle of incidence in degrees

- **Method:** sky = DHI [$(1 - A)$(1 + cos beta)/2 (1 + sqrt(BHI/GHI) sin^3(beta/2)) + A R_b]
- **Assumes:** sun above the horizon; DHI <= GHI; 0 <= albedo <= 1
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_poa_reindl', tilt=30, surface_azimuth=180, zenith=30,
azimuth=180, dni=800, ghi=792.820323027551, dhi=100, albedo=0.2, n=172)
# -> poa_global = 917.939 W/m^2, poa_direct = 800 W/m^2, poa_sky_diffuse = 107.317 W/m^2,
poa_ground_diffuse = 10.6218 W/m^2, aoi_deg = 0   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Plane of array irradiance (Reindl): tilt = 30, surface_azimuth = 180, zenith = 30, azimuth = 180, dni = 800 W/m^2, ghi = 792.820323027551 W/m^2, dhi = 100 W/m^2, albedo = 0.2, n = 172	SOLVED	poa_global = 917.939 W/m^2, poa_direct = 800 W/m^2, poa_sky_diffuse = 107.317 W/m^2, poa_ground_diffuse = 10.6218 W/m^2, aoi_deg = 0
Plane of array irradiance (Reindl): tilt = 30, surface_azimuth = 180, zenith = 30, azimuth = 180, dni = 800 W/m^2, ghi = 792.820323027551 W/m^2, dhi = 100 W/m^2, albedo = 1.5, n = 172	INVALID_INPUT	Irradiance on a tilted plane (Reindl sky model): the albedo must lie between 0 and 1.
Plane of array irradiance (Reindl): tilt = 30, surface_azimuth = 180, zenith = 30, azimuth = 180, dni = 800 W/m^2, ghi = 792.820323027551 W/m^2, dhi = 100 W/m^2, n = 172	NEEDS_INPUT	energy_poa_reindl needs albedo (ground reflectance (0 to 1)). Example: solve('energy_poa_reindl', tilt=30, surface_azimu

Diffuse and direct irradiance from global horizontal irradiance (Erbs) — `energy_erbs_decomposition`

The clearness index, diffuse fraction, diffuse horizontal and direct normal irradiance estimated from a measured global horizontal irradiance by the Erbs (1982) correlation.

$$\frac{G_d}{G} = 0.9511 - 0.1604k_t + 4.388k_t^2 - 16.638k_t^3 + 12.336k_t^4$$

input	meaning	condition
`ghi`	global horizontal irradiance	
`zenith`	solar zenith angle in degrees	
`n`	day of the year (1 = 1 January)	

output	meaning
`kt`	clearness index GHI/(G_on cos z)

output	meaning
`diffuse_fraction`	diffuse fraction DHI/GHI
`dhi`	diffuse horizontal irradiance
`dni`	direct normal irradiance

- **Method:** kt = GHI/(G_on cos z); DHI/GHI = 1 - 0.09 kt $(kt \le 0.22)$, $0.9511 - 0.1604\,kt + 4.388\,kt^2 - 16.638\,kt^3 + 12.336\,kt^4$ (<= 0.8), 0.165; DNI = (GHI - DHI)/cos z
- **Assumes:** 0 <= zenith <= 85; GHI > 0; kt is capped at 1
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_erbs_decomposition', ghi=528.6494370285654, zenith=60, n=172)
# -> kt = 0.8, diffuse_fraction = 0.16527, dhi = 87.3697 W/m^2, dni = 882.56 W/m^2   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_erbs_decomposition', ghi=141.3981805, zenith=0, n=1)
# -> kt = 0.1, diffuse_fraction = 0.991, dhi = 140.126 W/m^2, dni = 1.27258 W/m^2   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Erbs decomposition: ghi = 141.3981805 W/m^2, zenith = 0, n = 1	SOLVED	kt = 0.1, diffuse_fraction = 0.991, dhi = 140.126 W/m^2, dni = 1.27258 W/m^2
Erbs decomposition: ghi = 600 W/m^2, zenith = 88, n = 172	INVALID_INPUT	Diffuse and direct irradiance from global horizontal irradiance (Erbs): the zenith angle must lie between 0 and 85 degre

Clear-sky global horizontal irradiance (Haurwitz) — `energy_clear_sky_haurwitz`

The global horizontal irradiance under a clear sky from the solar zenith angle alone (Haurwitz 1945): 1098 cos z exp($-0.059/cos$ z).

$$G_{cs} = 1098 \cos z \, e^{-0.059/\cos z}$$

input	meaning	condition
`zenith`	apparent solar zenith angle in degrees	

output	meaning
`ghi`	clear-sky global horizontal irradiance (0 when the sun is below the horizon)

- **Method:** GHI = 1098 cos z exp($-0.059/cos$ z) for cos $z > 0$, else 0
- **Assumes:** 0 <= zenith <= 180
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_clear_sky_haurwitz', zenith=0)
# -> ghi = 1035.09 W/m^2   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_clear_sky_haurwitz', zenith=95)
# -> ghi = 0 W/m^2   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Clear-sky GHI (Haurwitz): zenith = 0	SOLVED	ghi = 1035.09 W/m^2
Haurwitz clear sky: zenith = 200	INVALID_INPUT	Clear-sky global horizontal irradiance (Haurwitz): the zenith angle must lie between 0 and 180 degrees.

PV cell temperature from NOCT — `energy_cell_temperature_noct`

The operating temperature of a PV cell from the plane-of-array irradiance, the air temperature and the module's nominal operating cell temperature (NOCT): T_c = T_a + (NOCT - 20 degC) $G/800$.

$$T_c = T_a + \frac{NOCT - 20°\mathrm{C}}{800\ \mathrm{W/m^2}}\,G$$

input	meaning	condition
`poa`	plane-of-array irradiance	
`T_air`	ambient air temperature	
`noct`	nominal operating cell temperature (at 800 W/m^2, 20 degC, 1 m/s)	

output	meaning
`T_cell`	cell temperature
`T_cell_C`	cell temperature in degC

- **Method:** T_c = T_a + (NOCT - 20)/800 G (Ross model with the NOCT coefficient)
- **Assumes:** irradiance >= 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_cell_temperature_noct', poa=800, T_air=298.15, noct=318.15)
# -> T_cell = 323.15 K, T_cell_C = 50   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_cell_temperature_noct', poa=1000, T_air=293.15, noct=320.15)
# -> T_cell = 326.9 K, T_cell_C = 53.75   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Cell temperature (NOCT): poa = 800 W/m^2, T_air = 25 degC, noct = 45 degC	SOLVED	T_cell = 323.15 K, T_cell_C = 50
Cell temperature (NOCT): poa = -800 W/m^2, T_air = 25 degC, noct = 45 degC	INVALID_INPUT	PV cell temperature from NOCT: the plane-of-array irradiance must be zero or positive.

PV module temperature (Faiman model) — `energy_cell_temperature_faiman`

Module (cell) temperature by the Faiman heat-loss model used in IEC 61853: T = T_a + G/(u0 + u1 v).

$$T_m = T_a + \frac{G}{u_0 + u_1 v}$$

input	meaning	condition
`poa`	plane-of-array irradiance	
`T_air`	ambient air temperature	
`wind`	wind speed at module height	
`u0`	constant heat-loss coefficient in W/(m^2 K) (default 25)	
`u1`	wind heat-loss coefficient in W s/(m^3 K) (default 6.84)	

output	meaning
`T_cell`	module temperature
`T_cell_C`	module temperature in degC

- **Method:** T = T_a + G/(u0 + u1 v)
- **Assumes:** $G \geq 0$; $u_0 > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_cell_temperature_faiman', poa=1000, T_air=298.15, wind=0)
# -> T_cell = 338.15 K, T_cell_C = 65   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_cell_temperature_faiman', poa=800, T_air=293.15, wind=1, u0=30,
u1=10)
# -> T_cell = 313.15 K, T_cell_C = 40   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Module temperature (Faiman): poa = 1000 W/m^2, T_air = 25 degC, wind = 0 m/s	SOLVED	T_cell = 338.15 K, T_cell_C = 65
Faiman temperature: poa = 800 W/m^2, T_air = 20 degC, wind = 1 m/s, u0 = 30, u1 = 10	SOLVED	T_cell = 313.15 K, T_cell_C = 40
Module temperature (Faiman): poa = 1000 W/m^2, T_air = 25 degC	NEEDS_INPUT	energy_cell_temperature_faiman needs wind (wind speed at module height). Example: solve('energy_cell_temperature_faiman'

PV cell and module temperature (Sandia SAPM model) — `energy_cell_temperature_sapm`

Back-of-module and cell temperature by the Sandia Array Performance Model, $T_m = G$ exp(a + b v) + T_a, $T_c = T_m + (G/1000)$ dT, for the standard module/mounting parameter sets.

$$T_m = G\,e^{a+bv} + T_a, \quad T_c = T_m + \frac{G}{1000}\Delta T$$

input	meaning	condition
`poa`	plane-of-array irradiance	
`T_air`	ambient air temperature	
`wind`	wind speed at 10 m	
`mount`	module construction and mounting (Sandia parameter set)	

output	meaning
`T_module_C`	back-of-module temperature in degC
`T_cell_C`	cell temperature in degC
`T_cell`	cell temperature

- **Method:** $T_m = G$ exp(a + b v) + T_a; $T_c = T_m + G\ dT/1000$ (a, b, dT from the Sandia table)
- **Assumes:** $G \geq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_cell_temperature_sapm', poa=1000, T_air=283.15, wind=0,
mount='open_rack_glass_glass')
# -> T_module_C = 41.117, T_cell_C = 44.117, T_cell = 317.267 K   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Cell temperature (SAPM): poa = 1000 W/m^2, T_air = 10 degC, wind = 0 m/s, mount = open_rack_glass_glass	SOLVED	T_module_C = 41.117, T_cell_C = 44.117, T_cell = 317.267 K
Cell temperature (SAPM): poa = 1000 W/m^2, T_air = 10 degC, wind = 0 m/s, mount = on_the_moon	INVALID_INPUT	mount: one of open_rack_glass_glass, close_mount_glass_glass, open_rack_glass_polymer, insulated_back_glass_polymer; got

PV array DC power with temperature derating (PVWatts) — `energy_pvwatts_dc`

DC power of a PV array from the effective irradiance and cell temperature (PVWatts): $P = (G/1000)$ P_dc0 (1 + gamma (T_c - 25 degC)).

$$P_{dc} = \frac{G}{1000} P_{dc0}\,[1 + \gamma(T_c - 25°\mathrm{C})]$$

input	meaning	condition
`G_eff`	effective (transmitted) irradiance on the cells	
`T_cell`	cell temperature	
`pdc0`	DC rating at 1000 W/m^2 and 25 degC	
`gamma`	power temperature coefficient per K (e.g. -0.004, $or -0.4\%$)	

output	meaning
`pdc`	DC power
`pdc_kW`	DC power in kW

- **Method:** $P_dc = (G/1000)$ P_dc0 [$1 + \gamma\,(T_c - 25)$]
- **Assumes:** $-0.02 \le \gamma \le 0.01$ per K
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_pvwatts_dc', G_eff=800, T_cell=318.15, pdc0=5000, gamma=-0.004)
# -> pdc = 3680 W, pdc_kW = 3.68   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_pvwatts_dc', G_eff=1000, T_cell=298.15, pdc0=300,
gamma=-0.0035)
# -> pdc = 300 W, pdc_kW = 0.3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
PVWatts DC power: G_eff = 800 W/m^2, T_cell = 45 degC, pdc0 = 5 kW, gamma = -0.4%	SOLVED	pdc = 3680 W, pdc_kW = 3.68
PVWatts DC power: G_eff = 800 W/m^2, T_cell = 45 degC, pdc0 = 5 kW, gamma = -0.4	INVALID_INPUT	PV array DC power with temperature derating (PVWatts): the temperature coefficient gamma must lie between -0.02 and 0.01

PV array output at operating temperature — `energy_pv_array_output`

Cell temperature from NOCT and the resulting DC output of an array with a power temperature coefficient, from the plane-of-array irradiance and the air temperature.

$$P = \frac{G}{1000} P_{dc0} \left[1 + \gamma \left(T_a + \frac{NOCT - 20}{800} G - 25\right)\right]$$

input	meaning	condition
`poa`	plane-of-array irradiance	
`T_air`	ambient air temperature	
`noct`	nominal operating cell temperature	
`pdc0`	array DC rating at STC	
`gamma`	power temperature coefficient per K (e.g. -0.004, $or -0.4$%)	

output	meaning
`T_cell`	cell temperature
`T_cell_C`	cell temperature in degC
`pdc`	DC output
`derate`	temperature derating factor $1 + \gamma\,(T_c - 25)$

- **Method:** T_c = T_a + (NOCT - 20) $G/800$; $P = (G/1000)$ P_dc0 [$1 + \gamma\,(T_c - 25)$]
- **Assumes:** $G > 0$; $-0.02 \le \gamma \le 0.01$ per K
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_pv_array_output', poa=800, T_air=298.15, noct=318.15,
pdc0=10000, gamma=-0.004)
# -> T_cell = 323.15 K, T_cell_C = 50, pdc = 7200 W, derate = 0.9   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
PV array output: poa = 800 W/m^2, T_air = 25 degC, noct = 45 degC, pdc0 = 10 kW, gamma = -0.4%	SOLVED	T_cell = 323.15 K, T_cell_C = 50, pdc = 7200 W, derate = 0.9
PV array output: poa = 0 W/m^2, T_air = 25 degC, noct = 45 degC, pdc0 = 10 kW, gamma = -0.4%	INVALID_INPUT	PV array output at operating temperature: the plane-of-array irradiance must be a positive number.

Inverter AC output and efficiency (PVWatts) — `energy_pvwatts_inverter`

AC output and efficiency of an inverter at a DC input by the PVWatts part-load efficiency curve, with clipping at the AC rating eta_nom P_dc0.

$$\eta = \frac{\eta_{nom}}{\eta_{ref}}\left(-0.0162\zeta - \frac{0.0059}{\zeta} + 0.9858\right),\ \zeta = \frac{P_{dc}}{P_{dc0}}$$

input	meaning	condition
`pdc`	DC input power	
`pdc0`	inverter DC input rating	
`eta_nom`	nominal efficiency (default 0.96)	
`eta_ref`	reference efficiency of the curve (default 0.9637)	

output	meaning
`pac`	AC output power
`efficiency`	conversion efficiency P_ac/P_dc
`clipped`	true when the output is limited at the AC rating

- **Method:** $\zeta = P_dc/P_dc0$; eta = (eta_nom/eta_ref)(-0.0162 zeta - 0.0059/zeta + 0.9858); P_ac = min(eta P_dc, eta_nom P_dc0)
- **Assumes:** $P_dc > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_pvwatts_inverter', pdc=5000, pdc0=5000)
# -> pac = 4800 W, efficiency = 0.96, clipped = True   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_pvwatts_inverter', pdc=6000, pdc0=5000)
# -> pac = 4800 W, efficiency = 0.8, clipped = True   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Inverter output (PVWatts): pdc = 6 kW, pdc0 = 5 kW	SOLVED	pac = 4800 W, efficiency = 0.8, clipped = True
Inverter output (PVWatts): pdc = 0 kW, pdc0 = 5 kW	INVALID_INPUT	Inverter AC output and efficiency (PVWatts): the DC input power must be a positive number.

Incidence angle modifier (ASHRAE) — `energy_iam_ashrae`

The fraction of beam irradiance transmitted into a PV module at an angle of incidence, by the ASHRAE model IAM = 1 - b0 (1/cos(theta) - 1), floored at 0.

$$IAM = 1 - b_0 \left(\frac{1}{\cos \theta} - 1 \right)$$

input	meaning	condition
`aoi`	angle of incidence in degrees	
`b0`	ASHRAE coefficient (default 0.05)	

output	meaning
`iam`	incidence angle modifier

- **Method:** IAM = max(0, $1 - b_0$ (sec(theta) - 1)); 0 for $\theta \geq 90$
- **Assumes:** 0 <= aoi <= 180
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_iam_ashrae', aoi=60)
# -> iam = 0.95   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_iam_ashrae', aoi=0)
# -> iam = 1   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Incidence angle modifier (ASHRAE): aoi = 60	SOLVED	iam = 0.95
ASHRAE IAM: aoi = 60, b0 = 2	INVALID_INPUT	Incidence angle modifier (ASHRAE): b0 must lie between 0 and 1.

Incidence angle modifier from Fresnel reflection and absorption — `energy_iam_physical`

The incidence angle modifier of a glass cover from Fresnel reflection (both polarizations) and absorption in the glass (De Soto / physical model), normalised to normal incidence.

$$IAM = \frac{\frac{1}{2}[(1 - r_s) + (1 - r_p)]e^{-KL/\cos\theta_r}}{(1 - r_0)e^{-KL}}$$

input	meaning	condition
`aoi`	angle of incidence in degrees	
`n`	refractive index of the glass (default 1.526)	
`K`	glazing extinction coefficient in $1/m$ (default 4)	
`L`	glazing thickness (default 2 mm)	

output	meaning
`iam`	incidence angle modifier

- **Method:** Snell's law, Fresnel reflectances r_s, r_p, transmission $\tau = (1 - r)$ exp(-K L/cos theta_r), IAM = (tau_s + tau_p)/2/tau(0)
- **Assumes:** 0 <= aoi < 90; $1 < n < 3$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_iam_physical', aoi=0)
# -> iam = 1   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Incidence angle modifier (physical): aoi = 0	SOLVED	iam = 1
Incidence angle modifier (physical): aoi = 95	INVALID_INPUT	Incidence angle modifier from Fresnel reflection and absorption: the angle of incidence must lie between 0 and 90 degree

Combined PV system losses (PVWatts) — `energy_pvwatts_losses`

The total system loss of a PV system from its individual loss percentages, combined multiplicatively as in PVWatts: 1 - prod(1 - L_i); unstated losses take the PVWatts defaults (soiling 2%, shading 3%, snow 0%, mismatch 2%, wiring 2%, connections 0.5%, light-induced degradation 1.5%, nameplate 1%, age 0%, availability 3%).

$$L = 1 - \prod_i (1 - L_i)$$

input	meaning	condition
`soiling`	soiling loss as a fraction or a percentage (default 2.0%)	
`shading`	shading loss as a fraction or a percentage (default 3.0%)	

input	meaning	condition
`snow`	snow loss as a fraction or a percentage (default 0.0%)	
`mismatch`	mismatch loss as a fraction or a percentage (default 2.0%)	
`wiring`	wiring loss as a fraction or a percentage (default 2.0%)	
`connections`	connections loss as a fraction or a percentage (default 0.5%)	
`lid`	lid loss as a fraction or a percentage (default 1.5%)	
`nameplate_rating`	nameplate rating loss as a fraction or a percentage (default 1.0%)	
`age`	age loss as a fraction or a percentage (default 0.0%)	
`availability`	availability loss as a fraction or a percentage (default 3.0%)	

output	meaning
`total_loss`	total loss as a fraction
`performance_factor`	product of (1 - loss)
`total_loss_pct`	total loss in percent

- **Method:** L = 1 - prod(1 - L_i)
- **Assumes:** each loss between 0 and 100%
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_pvwatts_losses', )
# -> total_loss = 0.140757, performance_factor = 0.859243, total_loss_pct = 14.0757   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_pvwatts_losses', soiling=0.0, shading=0.0, snow=0.0,
mismatch=0.0, wiring=0.0, connections=0.0, lid=0.0, nameplate_rating=0.0, age=0.0,
availability=0.0)
# -> total_loss = 0, performance_factor = 1, total_loss_pct = 0   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
PVWatts losses: soiling = 0%, shading = 0%, snow = 0%, mismatch = 0%, wiring = 0%, connections = 0%, lid = 0%, nameplate_rating = 0%, age = 0%, availability = 10%	SOLVED	total_loss = 0.1, performance_factor = 0.9, total_loss_pct = 10
PV system losses: soiling = 120%	INVALID_INPUT	Combined PV system losses (PVWatts): the soiling loss must lie between 0 and 100%.

Power in the wind and the Betz limit — `energy_wind_power`

The kinetic power of the wind through a rotor's swept area, the maximum a turbine can extract (Betz limit 16/27), the power density, and the power at a given power coefficient.

$$P = \tfrac{1}{2}\rho A v^3, \quad P_{max} = \tfrac{16}{27}\cdot\tfrac{1}{2}\rho A v^3$$

input	meaning	condition
`rho`	air density	
`D`	rotor diameter	
`v`	wind speed	
`cp`	power coefficient (optional, at most 16/27)	

output	meaning
`P_wind`	power in the wind through the swept area
`P_betz`	maximum extractable power (Betz)
`power_density`	wind power density in W/m^2
`area`	swept area
`P`	power at the given power coefficient

- **Method:** $P = (1/2)$ rho A v^3, $A = pi\ D^2/4$; P_Betz = (16/27) P; P(cp) = cp P
- **Assumes:** rho, D, $v > 0$; $0 < cp \leq 16/27$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_wind_power', rho=1.2, D=1.1283791670955126, v=10)
# -> P_wind = 600 W, P_betz = 355.556 W, power_density = 600, area = 1 m^2   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_wind_power', rho=1.225, D=80, v=10, cp=0.4)
# -> P_wind = 3.07876e+06 W, P_betz = 1.82445e+06 W, power_density = 612.5, area = 5026.55 m^2, P =
1.2315e+06 W   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Power in the wind: rho = 1.225 kg/m^3, D = 80 m, v = 10 m/s, cp = 0.4	SOLVED	P_wind = 3.07876e+06 W, P_betz = 1.82445e+06 W, power_density = 612.5, area = 5026.55 m^2, P = 1.2315e+06 W
Wind power: rho = 1.2 kg/m^3, D = 100 m, v = 12 m/s, cp = 0.7	INVALID_INPUT	Power in the wind and the Betz limit: the power coefficient must lie between 0 and the Betz limit 16/27 = 0.593.
Wind power: rho = 1.2 kg/m^3, D = 100 m	NEEDS_INPUT	energy_wind_power needs v (wind speed). Example: solve('energy_wind_power', rho='1.2 kg/m^3', D='1.12838 m', v='10 m/s')

Actuator-disc (momentum theory) rotor performance — `energy_actuator_disc`

Power and thrust coefficients, disc and far-wake speeds, power and thrust of an ideal rotor (actuator disc) with axial induction factor a; $a = 1/3$ gives the Betz optimum $Cp = 16/27.$

$$C_P = 4a(1-a)^2, \quad C_T = 4a(1-a)$$

input	meaning	condition
`a`	axial induction factor $(0 \leq a < 0.5)$	

input	meaning	condition
`rho`	air density	
`D`	rotor diameter	
`v`	free-stream wind speed	

output	meaning
`Cp`	power coefficient 4a(1-a)^2
`Ct`	thrust coefficient 4a(1-a)
`v_disc`	speed at the disc
`v_wake`	far-wake speed
`P`	power extracted
`T`	thrust on the rotor

- **Method:** $Cp = 4a$(1 - a)^2, $Ct = 4a$(1 - a), v_disc = v(1 - a), v_wake = v(1 - 2a); $P = Cp\,(1/2)$ rho A v^3, $T = Ct\,(1/2)$ rho A v^2
- **Assumes:** $0 \le a < 0.5$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_actuator_disc', a=0.3333333333333333, rho=1.2, D=10, v=8)
# -> Cp = 0.592593, Ct = 0.888889, v_disc = 5.33333 m/s, v_wake = 2.66667 m/s, P = 14297.7 W, T =
2680.83 N   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_actuator_disc', a=0.25, rho=1.2, D=10, v=8)
# -> Cp = 0.5625, Ct = 0.75, v_disc = 6 m/s, v_wake = 4 m/s, P = 13571.7 W, T = 2261.95 N
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Actuator disc: a = 0.25, rho = 1.2 kg/m^3, D = 10 m, v = 8 m/s	SOLVED	Cp = 0.5625, Ct = 0.75, v_disc = 6 m/s, v_wake = 4 m/s, P = 13571.7 W, T = 2261.95 N
Actuator disc: a = 0.6, rho = 1.2 kg/m^3, D = 10 m, v = 8 m/s	INVALID_INPUT	Actuator-disc (momentum theory) rotor performance: the axial induction factor a must lie in [0, 0.5) (momentum theory fa

Wind speed at another height (power law) — `energy_wind_shear_power_law`

The wind speed at height h2 from a measurement at h1 by the power (Hellman) law $v_2 = v_1\,(h_2/h_1)^{\alpha}$; $\alpha = 1/7$ when not stated (the one-seventh law).

$$v_2 = v_1 \left(\frac{h_2}{h_1} \right)^{\alpha}$$

input	meaning	condition
`v1`	measured wind speed	
`h1`	measurement height	

input	meaning	condition
`h2`	target height (e.g. hub height)	
`alpha`	shear exponent (default 1/7)	

output	meaning
`v2`	wind speed at h2

- **Method:** $v_2 = v_1\,(h_2/h_1)^{\alpha}$
- **Assumes:** heights > 0; $0 \le \alpha \le 1$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_wind_shear_power_law', v1=5, h1=10, h2=80,
alpha=0.3333333333333333)
# -> v2 = 10 m/s   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_wind_shear_power_law', v1=6, h1=10, h2=1280)
# -> v2 = 12 m/s   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Wind shear (power law): v1 = 6 m/s, h1 = 10 m, h2 = 1280 m	SOLVED	v2 = 12 m/s
Wind shear (power law): v1 = 5 m/s, h1 = 10 m, h2 = 80 m, alpha = 0.3333333333333333	SOLVED	v2 = 10 m/s
Wind shear (power law): v1 = 5 m/s, h1 = 0 m, h2 = 80 m	INVALID_INPUT	Wind speed at another height (power law): the wind speed and the heights must be a positive number.

Wind speed at another height (logarithmic profile) — `energy_wind_shear_log`

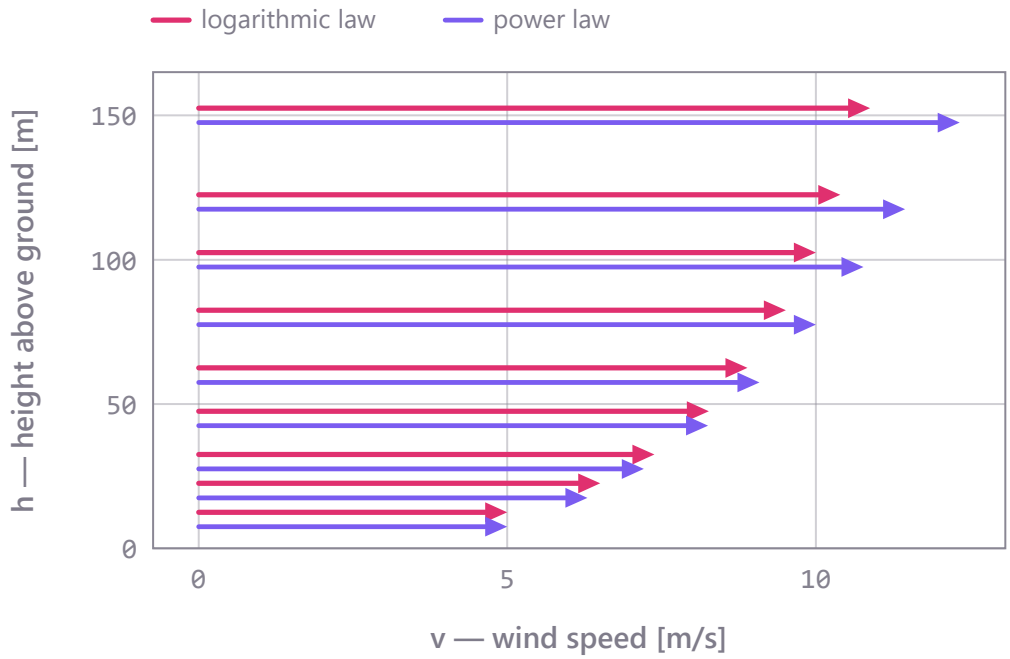
The wind speed at height h2 from a measurement at h1 by the logarithmic wind profile with roughness length z0 and an optional obstacle height d (displacement 0.7 d).

$$v_2 = v_1\,\frac{\ln\left((h_2 - 0.7d)/z_0\right)}{\ln\left((h_1 - 0.7d)/z_0\right)}$$

input	meaning	condition
`v1`	measured wind speed	
`h1`	measurement height	
`h2`	target height	
`z0`	surface roughness length	
`d`	obstacle height (default 0; displacement height 0.7 d)	

output	meaning
`v2`	wind speed at h2

- **Method:** $v_2 = v_1$ ln((h2 - 0.7 d)/z0)/ln((h1 - 0.7 d)/z0)
- **Assumes:** $h - 0.7\,d > z_0$ at both heights
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Wind speed at nine heights extrapolated from 5 m/s measured at 10 m, drawn as arrows whose length is the speed: logarithmic law (roughness length 1 m) and power law (exponent 0.333333), each computed by the package. At 100 m: 10 and 10.7722 m/s; at 150 m: 10.8805 and 12.3311 m/s.

```
energy_systems_ai999x.solve('energy_wind_shear_log', v1=5, h1=10, h2=100, z0=1)
# -> v2 = 10 m/s   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_wind_shear_log', v1=4, h1=9.718281828459045,
h2=27.085536923187664, z0=1, d=10)
# -> v2 = 12 m/s   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Wind shear (log profile): v1 = 5 m/s, h1 = 10 m, h2 = 100 m, z0 = 1 m	SOLVED	v2 = 10 m/s
Wind shear (log profile): v1 = 5 m/s, h1 = 10 m, h2 = 100 m, z0 = 20 m	INVALID_INPUT	Wind speed at another height (logarithmic profile): both heights must exceed the displacement height 0.7 d plus the roug

Air density at hub height — `energy_hub_air_density`

Temperature, pressure and air density at a turbine's hub height from a pressure and a temperature measured at other heights: temperature gradient -6.5 K/km, pressure gradient -1/8 hPa per m, then the barometric (1.225 kg/m^3 at 288.15 K, 1013.3 hPa) or the ideal-gas (R = 287.058 J/(kg K)) density.

$$\rho_{hub} = 1.225\,\frac{p_{hub}}{101330\ \mathrm{Pa}}\,\frac{288.15\ \mathrm{K}}{T_{hub}}$$

input	meaning	condition
`p`	measured air pressure	
`h_p`	height of the pressure measurement	
`T`	measured air temperature	
`h_T`	height of the temperature measurement	
`hub`	hub height	
`model`	density model (default barometric)	

output	meaning
`T_hub`	temperature at hub height
`p_hub`	pressure at hub height
`rho`	air density at hub height

- **Method:** T_hub = T - 0.0065 (h_hub - h_T); p_hub = p - 12.5 Pa/m (h_hub - h_p); rho = p_hub/(R T_hub) (ideal gas) or 1.225 (p_hub/101330)(288.15/T_hub) (barometric)
- **Assumes:** pressure and temperature stay positive at hub height
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_hub_air_density', p=101330, h_p=0, T=288.15, h_T=0, hub=0)
# -> T_hub = 288.15 K, p_hub = 101330 Pa, rho = 1.225 kg/m^3   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_hub_air_density', p=101330, h_p=0, T=288.15, h_T=0, hub=0,
model='ideal_gas')
# -> T_hub = 288.15 K, p_hub = 101330 Pa, rho = 1.22504 kg/m^3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Air density at hub height: p = 1013.3 hPa, h_p = 0 m, T = 288.15 K, h_T = 0 m, hub = 0 m	SOLVED	T_hub = 288.15 K, p_hub = 101330 Pa, rho = 1.225 kg/m^3
Air density at hub height: p = 1013 hPa, h_p = 0 m, T = 15 degC, h_T = 2 m	NEEDS_INPUT	energy_hub_air_density needs hub (hub height). Example: solve('energy_hub_air_density', p='101330 Pa', h_p='0 m', T='288

Turbine output from its power curve — `energy_power_curve_output`

The electrical output of a wind turbine at a wind speed by linear interpolation of its power curve (zero below the first and above the last listed speed).

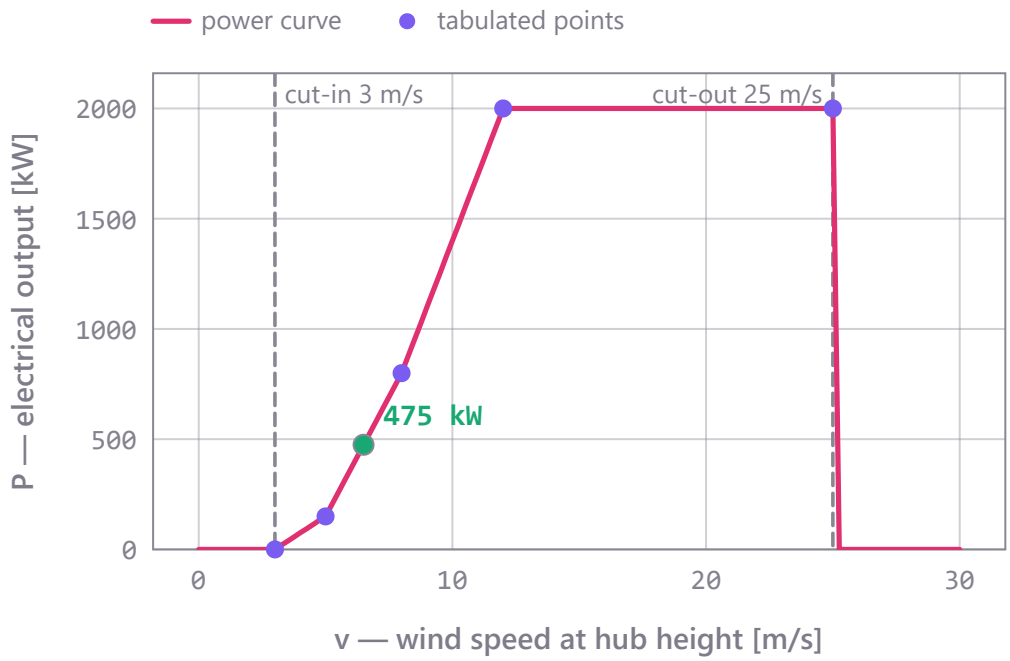
$$P(v) = P_i + (P_{i+1} - P_i)\frac{v - v_i}{v_{i+1} - v_i}$$

input	meaning	condition
`speeds`	power-curve wind speeds in m/s (increasing)	
`powers`	power-curve outputs in kW	

input	meaning	condition
`v`	wind speed at hub height	

output	meaning
`P_kW`	output in kW
`P`	output

- **Method:** linear interpolation between the curve points; 0 outside the listed speeds
- **Assumes:** speeds strictly increasing; one power per speed
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Wind turbine output against wind speed from the tabulated power curve (speeds 3, 5, 8, 12, 25 m/s, powers 0, 150, 800, 2000, 2000 kW), computed by the package at 121 speeds from 0 to 30 m/s: nothing below cut-in at 3 m/s, rated 2000 kW up to cut-out at 25 m/s, nothing above. Marked: v = 6.5 m/s, P = 475 kW.

```
energy_systems_ai999x.solve('energy_power_curve_output', speeds='3, 5, 8, 12, 25', powers='0, 150, 800, 2000, 2000', v=6.5)
# -> P_kW = 475, P = 475000 W   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_power_curve_output', speeds=[3, 5, 8, 12, 25], powers=[0, 150, 800, 2000, 2000], v=26)
# -> P_kW = 0, P = 0 W   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Power curve output: speeds = [3, 5, 8, 12, 25], powers = [0, 150, 800, 2000, 2000], v = 6.5 m/s	SOLVED	P_kW = 475, P = 475000 W
Power curve output: speeds = [3, 5, 4, 12, 25], powers = [0, 150, 800, 2000, 2000], v = 6.5 m/s	INVALID_INPUT	Turbine output from its power curve: the wind speeds of the curve must increase strictly.

question	outcome	result
Power curve output: speeds = [3, 5, 8, 12, 25], powers = [0, 150, 800, 2000, 2000]	NEEDS_INPUT	energy_power_curve_output needs v (wind speed at hub height). Example: solve('energy_power_curve_output', speeds='3, 5,

Turbine output from a density-corrected power curve — `energy_power_curve_density_corrected`

Turbine output at a wind speed and air density, with the power curve's speeds rescaled by $(1.225/\rho)^e$, e rising from $1/3$ below 7.5 m/s to $2/3$ above 12.5 m/s (Svenningsen's correction, as in windpowerlib).

$$v_i' = v_i \left(\frac{1.225}{\rho} \right)^{e(v_i)},\ e \in [\tfrac{1}{3}, \tfrac{2}{3}]$$

input	meaning	condition
`speeds`	power-curve wind speeds in m/s at 1.225 kg/m^3 (increasing)	
`powers`	power-curve outputs in kW	
`v`	wind speed at hub height	
`rho`	air density at hub height	

output	meaning
`P_kW`	output in kW
`P`	output

- **Method:** v_i' = v_i $(1.225/\rho)^e$(v_i); interpolate P(v) on (v_i', P_i)
- **Assumes:** $0.5 < \rho < 1.6\ kg/m^3$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_power_curve_density_corrected', speeds=[3, 5, 8, 12, 25],
powers=[0, 150, 800, 2000, 2000], v=6.5, rho=1.225)
# -> P_kW = 475, P = 475000 W   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Density-corrected power curve: speeds = [3, 5, 8, 12, 25], powers = [0, 150, 800, 2000, 2000], v = 6.5 m/s, rho = 1.225 kg/m^3	SOLVED	P_kW = 475, P = 475000 W
Density-corrected power curve: speeds = [3, 5, 8, 12, 25], powers = [0, 150, 800, 2000, 2000], v = 6.5 m/s, rho = 3 kg/m^3	INVALID_INPUT	Turbine output from a density-corrected power curve: the air density must lie between 0.5 and 1.6 kg/m^3.

Weibull wind statistics and mean power density — `energy_weibull_wind`

Mean wind speed, mean wind power density, energy pattern factor, most probable speed and the speed carrying most energy for a Weibull wind-speed distribution with shape k and scale c.

$$\bar{v} = c\,\Gamma\!\left(1+\tfrac{1}{k}\right), \quad \frac{\bar{P}}{A} = \tfrac{1}{2}\rho c^3\,\Gamma\!\left(1+\tfrac{3}{k}\right)$$

input	meaning	condition
`k`	Weibull shape factor $(1 < k \le 5)$	
`c`	Weibull scale factor	
`rho`	air density	

output	meaning
`v_mean`	mean wind speed c Gamma(1 + 1/k)
`power_density`	mean wind power density in W/m^2
`energy_pattern_factor`	mean of v^3 over the cube of the mean
`v_most_probable`	most probable wind speed
`v_max_energy`	wind speed carrying the most energy

- **Method:** v_mean = c Gamma(1 + 1/k); $P/A = (1/2)$ rho c^3 Gamma(1 + 3/k); $v_m p = c$ ((k - 1)/k)^(1/k); v_maxE = c ((k + 2)/k)^(1/k)
- **Assumes:** $1 < k \le 5$; $c > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_weibull_wind', k=2, c=8, rho=1.225)
# -> v_mean = 7.08982 m/s, power_density = 416.881, energy_pattern_factor = 1.90986,
v_most_probable = 5.65685 m/s, v_max_energy = 11.3137 m/s   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Weibull wind statistics: k = 2, c = 8 m/s, rho = 1.225 kg/m^3	SOLVED	v_mean = 7.08982 m/s, power_density = 416.881, energy_pattern_factor = 1.90986, v_most_probable = 5.65685 m/s, v_max_energy = 11.3137 m/s
Weibull wind statistics: k = 0.5, c = 8 m/s, rho = 1.225 kg/m^3	INVALID_INPUT	Weibull wind statistics and mean power density: the Weibull shape k must lie in (1, 5].

Annual energy production from a power curve and a Weibull distribution — `energy_weibull_aep`

Mean output, annual energy production and capacity factor of a wind turbine whose power curve (linear between points, zero outside) operates in winds with a Weibull distribution (shape k, scale c).

$$\bar{P} = \int_0^\infty P(v)\,\frac{k}{c}\left(\frac{v}{c}\right)^{k-1} e^{-(v/c)^k}\,dv, \quad AEP = 8760\,\mathrm{h}\cdot\bar{P}$$

input	meaning	condition
`speeds`	power-curve wind speeds in m/s (increasing)	
`powers`	power-curve outputs in kW	
`k`	Weibull shape factor $(1 < k \le 5)$	

input	meaning	condition
`c`	Weibull scale factor	
`hours`	hours in the period (default 8760)	

output	meaning
`mean_power_kW`	mean output in kW
`AEP_MWh`	energy over the period in MWh
`capacity_factor`	mean output over the largest output of the curve

- **Method:** P_mean = sum over segments of integral (P_i + s (v - v_i)) f(v) dv, each in closed form: F(b) - F(a) and c Gamma(1 + 1/k) [P($1 + 1/k$, (b/c)^k) - P($1 + 1/k$, (a/c)^k)]
- **Assumes:** $1 < k \leq 5$; speeds strictly increasing
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_weibull_aep', speeds=[0, 100], powers=[1000, 1000], k=2, c=7)
# -> mean_power_kW = 1000, AEP_MWh = 8760, capacity_factor = 1   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_weibull_aep', speeds=[0, 1000], powers=[0, 1000], k=2, c=7)
# -> mean_power_kW = 6.20359, AEP_MWh = 54.3434, capacity_factor = 0.00620359   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Annual energy production: speeds = [0, 1000], powers = [0, 1000], k = 2, c = 7 m/s	SOLVED	mean_power_kW = 6.20359, AEP_MWh = 54.3434, capacity_factor = 0.00620359
Annual energy production: speeds = [3, 5, 8, 12, 25], powers = [0, 150, 800, 2000, 2000], k = 7, c = 7 m/s	INVALID_INPUT	Annual energy production from a power curve and a Weibull distribution: the Weibull shape k must lie in (1, 5].

Balanced three-phase power from line quantities — `energy_three_phase_power`

Real, reactive and apparent power of a balanced three-phase load from the line voltage, line current and power factor, with the phase voltage and phase current of a star or delta connection.

$$P = \sqrt{3}\, V_L I_L \cos\phi, \quad Q = \sqrt{3}\, V_L I_L \sin\phi$$

input	meaning	condition
`VL`	line-to-line rms voltage	
`IL`	line current (rms)	
`pf`	power factor $(0 < pf \leq 1)$	
`connection`	connection of the load (star = wye)	
`pf_type`	lagging (inductive, default) or leading (capacitive)	

output	meaning
`P`	real power (three-phase total)

output	meaning
`Q`	reactive power in var (positive lagging)
`S`	apparent power in VA
`V_phase`	voltage across each phase of the load
`I_phase`	current in each phase of the load

- **Method:** $S = sqrt$(3) V_L I_L, $P = S$ pf, $Q = S$ sin(acos pf); star: $V_{ph} = \frac{V_L}{\sqrt{3}}$, $I_ph = I_L$; delta: $V_ph = V_L$, $I_{ph} = \frac{I_L}{\sqrt{3}}$
- **Assumes:** balanced load; $0 < pf \leq 1$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_three_phase_power', VL=400, IL=50, pf=0.8, connection='star')
# -> P = 27712.8 W, Q = 20784.6 var, S = 34641 VA, V_phase = 230.94 V, I_phase = 50 A   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_three_phase_power', VL=400, IL=17.32050807568877, pf=1.0,
connection='delta')
# -> P = 12000 W, Q = 0 var, S = 12000 VA, V_phase = 400 V, I_phase = 10 A   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Three-phase power: VL = 400 V, IL = 50 A, pf = 0.8, connection = star	SOLVED	P = 27712.8 W, Q = 20784.6 var, S = 34641 VA, V_phase = 230.94 V, I_phase = 50 A
Three-phase power: VL = 400 V, IL = 50 A, pf = 1.2, connection = star	INVALID_INPUT	Balanced three-phase power from line quantities: the power factor must lie in (0, 1].
Three-phase power: VL = 400 V, IL = 50 A, pf = 0.8	NEEDS_INPUT	energy_three_phase_power needs connection (connection of the load (star = wye)). Example: solve('energy_three_phase_powe

Per-unit impedance on a new base — `energy_per_unit_change_of_base`

An impedance given in per unit on equipment ratings (S_old, V_old) converted to the per-unit system of the network (S_new, V_new).

$$Z_{pu,new} = Z_{pu,old}\left(\frac{V_{old}}{V_{new}}\right)^2 \frac{S_{new}}{S_{old}}$$

input	meaning	condition
`R_pu`	per-unit resistance on the old base	
`X_pu`	per-unit reactance on the old base	
`S_old`	old base (equipment rating) in VA	
`V_old`	old base voltage (line-to-line)	
`S_new`	new (system) base power in VA	
`V_new`	new base voltage (line-to-line)	

output	meaning
`R_pu_new`	per-unit resistance on the new base
`X_pu_new`	per-unit reactance on the new base
`Z_pu_new`	per-unit impedance magnitude on the new base

- **Method:** Z_new = Z_old (V_old/V_new)^2 (S_new/S_old)
- **Assumes:** $X_pu > 0$; positive bases
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_per_unit_change_of_base', R_pu=0.0, X_pu=0.1, S_old=50000000.0,
V_old=132000.0, S_new=100000000.0, V_new=138000.0)
# -> R_pu_new = 0, X_pu_new = 0.182987, Z_pu_new = 0.182987   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_per_unit_change_of_base', R_pu=0.01, X_pu=0.1,
S_old=10000000.0, V_old=11000.0, S_new=100000000.0, V_new=11000.0)
# -> R_pu_new = 0.1, X_pu_new = 1, Z_pu_new = 1.00499   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Per-unit change of base: R_pu = 0.01, X_pu = 0.1, S_old = 10 MVA, V_old = 11 kV, S_new = 100 MVA, V_new = 11 kV	SOLVED	R_pu_new = 0.1, X_pu_new = 1, Z_pu_new = 1.00499
Per-unit change of base: R_pu = 0.01, X_pu = 0.1, S_old = 10 MVA, V_old = 11 kV, S_new = 100 MVA	NEEDS_INPUT	energy_per_unit_change_of_base needs V_new (new base voltage (line-to-line)). Example: solve('energy_per_unit_change_of_

Short transmission line: sending-end voltage, regulation and efficiency — `energy_short_line`

Sending-end voltage and its angle, line current, losses, voltage regulation and efficiency of a short three-phase line (series R + jX, no shunt admittance) feeding a load at a known receiving-end voltage.

$$\mathbf{V}_s = \mathbf{V}_r + \mathbf{I}\,(R + jX), \quad \mathrm{VR} = \frac{|V_s| - |V_r|}{|V_r|}$$

input	meaning	condition
`Vr`	receiving-end line-to-line voltage	
`P`	load real power (three-phase)	
`pf`	load power factor	
`R`	line resistance per phase (total)	
`X`	line reactance per phase (total)	
`pf_type`	lagging (default) or leading load	

output	meaning
`Vs`	sending-end line voltage
`delta_deg`	angle of the sending voltage ahead of the receiving voltage, degrees

output	meaning
`I`	receiving-end line current
`losses`	line losses (three-phase)
`Ps`	sending-end real power
`regulation_pct`	voltage regulation in percent
`efficiency_pct`	transmission efficiency in percent

- **Method:** $V_s = V_r + I$ (R + jX) per phase, I = P/(sqrt(3) V_r pf) at angle -acos(pf); regulation = (V_s - V_r)/V_r; efficiency = P/(P + 3 I^2 R)
- **Assumes:** short line: shunt admittance neglected; balanced three-phase load
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_short_line', Vr=33000.0, P=10000000.0, pf=1.0, R=0.0, X=0.0)
# -> Vs = 33000 V, delta_deg = 0, I = 174.955 A, losses = 0 W, Ps = 1e+07 W, regulation_pct = 0,
efficiency_pct = 100   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_short_line', Vr=1732.0508075688772, P=300000, pf=1.0, R=1.0,
X=0.0)
# -> Vs = 1905.26 V, delta_deg = 0, I = 100 A, losses = 30000 W, Ps = 330000 W, regulation_pct =
10, efficiency_pct = 90.9091   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Short line: Vr = 33 kV, P = 10 MW, pf = 1, R = 0 ohm, X = 0 ohm	SOLVED	Vs = 33000 V, delta_deg = 0, I = 174.955 A, losses = 0 W, Ps = 1e+07 W, regulation_pct = 0, efficiency_pct = 100
Short line: Vr = 33 kV, P = 10 MW, pf = 0.8, R = 5 ohm, X = 20 ohm, pf_type = sideways	INVALID_INPUT	pf_type: one of lagging, leading; got 'sideways'.
Short line: Vr = 33 kV, P = 10 MW, pf = 0.8, R = 5 ohm	NEEDS_INPUT	energy_short_line needs X (line reactance per phase (total)). Example: solve('energy_short_line', Vr='33000 V', P='1e+07

Medium transmission line (nominal pi) — `energy_medium_line`

ABCD constants of the nominal-pi model and the sending-end voltage, current and power, losses, regulation and efficiency of a medium-length three-phase line feeding a load.

$$\begin{bmatrix} V_s \\ I_s \end{bmatrix} = \begin{bmatrix} 1 + \frac{ZY}{2} & Z \\ Y\left(1 + \frac{ZY}{4}\right) & 1 + \frac{ZY}{2} \end{bmatrix} \begin{bmatrix} V_r \\ I_r \end{bmatrix}$$

input	meaning	condition
`Vr`	receiving-end line-to-line voltage	
`P`	load real power (three-phase)	
`pf`	load power factor	
`R`	line resistance per phase (total)	

input	meaning	condition
`X`	line reactance per phase (total)	
`B`	total shunt susceptance of the line per phase	
`pf_type`	lagging (default) or leading load	

output	meaning
`Vs`	sending-end line voltage
`delta_deg`	angle of the sending voltage ahead of the receiving voltage, degrees
`Is`	sending-end current
`Ps`	sending-end real power
`losses`	line losses (three-phase)
`regulation_pct`	voltage regulation (V_s/|A| - V_r)/V_r in percent
`efficiency_pct`	transmission efficiency in percent
`A_mag`	|A| (= |D|)
`A_angle_deg`	angle of A in degrees
`B_mag`	|B| in ohms
`B_angle_deg`	angle of B in degrees

- **Method:** $A = D = 1 + ZY/2$, $B = Z$, $C = Y$(1 + ZY/4); $V_s = A$ V_r + B I_r, $I_s = C$ V_r + D I_r; regulation = (V_s/|A| - V_r)/V_r
- **Assumes:** nominal-pi model (half the shunt admittance at each end); balanced load
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_medium_line', Vr=132000.0, P=50000000.0, pf=1.0, R=0.0, X=40.0, B=0.0005)
# -> Vs = 131555 V, delta_deg = 6.61355, Is = 219.801 A, Ps = 5e+07 W, losses = 0 W, regulation_pct = 0.669903, efficiency_pct = 100, A_mag = 0.99, A_angle_deg = 0, B_mag = 40 ohm, B_angle_deg = 90 (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Medium line: Vr = 132 kV, P = 50 MW, pf = 1, R = 0 ohm, X = 40 ohm, B = 500 uS	SOLVED	Vs = 131555 V, delta_deg = 6.61355, Is = 219.801 A, Ps = 5e+07 W, losses = 0 W, regulation_pct = 0.669903, efficiency_pct = 100, A_mag = 0.99, A_angle_deg = 0, B_mag = 40 ohm, B_angle_deg = 90
Medium line: Vr = 132 kV, P = 50 MW, pf = 0.8, R = 10 ohm, X = 40 ohm	NEEDS_INPUT	energy_medium_line needs B (total shunt susceptance of the line per phase). Example: solve('energy_medium_line', Vr='132

Long transmission line (exact distributed parameters) — `energy_long_line`

Characteristic impedance, propagation constant, exact ABCD constants and the sending-end quantities, regulation and efficiency of a long three-phase line with uniformly distributed R, X and shunt B (the Ferranti rise appears as a negative regulation at light load).

$$A = \cosh \gamma l,\ B = Z_c \sinh \gamma l,\ C = \frac{\sinh \gamma l}{Z_c}, \quad Z_c = \sqrt{\frac{z}{y}},\ \gamma = \sqrt{zy}$$

input	meaning	condition
`Vr`	receiving-end line-to-line voltage	
`P`	load real power (three-phase)	
`pf`	load power factor	
`R`	line resistance per phase (total)	
`X`	line reactance per phase (total)	
`B`	total shunt susceptance of the line per phase	
`pf_type`	lagging (default) or leading load	

output	meaning
`Vs`	sending-end line voltage
`delta_deg`	angle of the sending voltage ahead of the receiving voltage, degrees
`Is`	sending-end current
`Ps`	sending-end real power
`losses`	line losses (three-phase)
`regulation_pct`	voltage regulation (V_s/|A| - V_r)/V_r in percent
`efficiency_pct`	transmission efficiency in percent
`A_mag`	|A| (= |D|)
`A_angle_deg`	angle of A in degrees
`B_mag`	|B| in ohms
`B_angle_deg`	angle of B in degrees
`Zc_mag`	characteristic (surge) impedance magnitude
`Zc_angle_deg`	angle of Zc in degrees
`alpha_l`	attenuation alpha l in nepers
`beta_l`	phase shift beta l in radians

- **Method:** gamma $l = \sqrt{ZY}$, $Zc = \sqrt{\frac{Z}{Y}}$; A = D = cosh(gamma l), $B = Zc$ sinh(gamma l), C = sinh(gamma l)/Zc
- **Assumes:** uniformly distributed parameters; balanced load
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_long_line', Vr=400000.0, P=1000000.0, pf=1.0, R=0.0, X=100.0,
B=0.0004)
# -> Vs = 392027 V, delta_deg = 0.0362951, Is = 91.7723 A, Ps = 1e+06 W, losses = 0 W,
regulation_pct = 2.00641e-05, efficiency_pct = 100, A_mag = 0.980067, A_angle_deg = 0, B_mag =
99.3347 ohm, B_angle_deg = 90, Zc_mag = 500 ohm, Zc_angle_deg = 0, alpha_l = 0, beta_l = 0.2
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Long line: Vr = 400 kV, P = 1 MW, pf = 1, R = 0 ohm, X = 100 ohm, B = 400 uS	SOLVED	Vs = 392027 V, delta_deg = 0.0362951, Is = 91.7723 A, Ps = 1e+06 W, losses = 0 W, regulation_pct = 2.00641e-05, efficiency_pct = 100, A_mag = 0.980067, A_angle_deg = 0, B_mag = 99.3347 ohm, B_angle_deg = 90, Zc_mag = 500 ohm, Zc_angle_deg = 0, alpha_l = 0, beta_l = 0.2
Long line: Vr = 400 kV, P = 1 MW, pf = 1, R = 0 ohm, X = 100 ohm, B = 0 uS	INVALID_INPUT	Long transmission line (exact distributed parameters): the shunt susceptance must be a positive number.

Inductance, capacitance and surge impedance of a three-phase overhead line — `energy_line_parameters`

Per-phase inductance, capacitance, reactance, susceptance, surge impedance (and surge-impedance loading at a voltage) of a fully transposed three-phase line from its phase spacings, conductor radius and geometric mean radius, with optional bundled conductors.

$$L = 2\times10^{-7}\ln\frac{GMD}{GMR}\ \mathrm{H/m},\quad C = \frac{2\pi\varepsilon_0}{\ln(GMD/r)}\ \mathrm{F/m}$$

input	meaning	condition
`D_ab`	distance between phases a and b	
`D_bc`	distance between phases b and c	
`D_ca`	distance between phases c and a	
`r`	conductor (outer) radius	
`f`	system frequency	
`gmr`	conductor geometric mean radius (default r $e^{-1/4}$, a solid round conductor)	
`n_bundle`	subconductors per phase (1 to 4, regular polygon)	
`spacing`	distance between neighbouring subconductors of a bundle	
`V`	line voltage for the surge-impedance loading (optional)	

output	meaning
`GMR`	geometric mean radius of a phase (bundle)
`L_mH_per_km`	inductance per phase in mH/km
`C_nF_per_km`	capacitance to neutral in nF/km
`X_ohm_per_km`	series reactance in ohm/km
`B_uS_per_km`	shunt susceptance in uS/km
`Zc`	surge impedance of the lossless line sqrt(L/C)
`SIL_MW`	surge-impedance loading V^2/Zc in MW

- **Method:** GMD = (D_ab D_bc D_ca)^(1/3); $L = 2e-7$ ln(GMD/GMR) H/m; $C = 2$ pi eps0/ln(GMD/r) F/m; bundles: GMR_b = (GMR d^{n-1})^$(1/n)$ (2^(1/8) (GMR d^3)^(1/4) = 1.09 (GMR d^3)^(1/4) for 4)
- **Assumes:** fully transposed line; spacings much larger than the conductors; ground effect neglected
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_line_parameters', D_ab=5, D_bc=5, D_ca=5, r=0.01, f=50,
gmr=0.005)
# -> GMR = 0.005 m, L_mH_per_km = 1.38155, C_nF_per_km = 8.95189, X_ohm_per_km = 0.434027,
B_uS_per_km = 2.81232, Zc = 392.849 ohm   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_line_parameters', D_ab=6, D_bc=6, D_ca=6, r=0.012, f=60,
gmr=0.01, n_bundle=2, spacing=0.4)
# -> GMR = 0.0632456 m, L_mH_per_km = 0.910498, C_nF_per_km = 12.4699, X_ohm_per_km = 0.34325,
B_uS_per_km = 4.70106, Zc = 270.214 ohm   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Line parameters: D_ab = 5 m, D_bc = 5 m, D_ca = 5 m, r = 1 cm, f = 50 Hz, gmr = 5 mm	SOLVED	GMR = 0.005 m, L_mH_per_km = 1.38155, C_nF_per_km = 8.95189, X_ohm_per_km = 0.434027, B_uS_per_km = 2.81232, Zc = 392.849 ohm
Line parameters: D_ab = 5 m, D_bc = 5 m, D_ca = 5 m, r = 1 cm, f = 50 Hz, gmr = 2 cm	INVALID_INPUT	Inductance, capacitance and surge impedance of a three-phase overhead line: the geometric mean radius cannot exceed the

Receiving-end voltage of a line feeding a constant-power load (two-bus load flow) — `energy_two_bus_load_flow`

The receiving-end voltage magnitude and angle, current and losses when a three-phase line of impedance R + jX supplies a constant P + jQ load from a fixed sending-end voltage (the exact two-bus load flow).

$$V_r^4 + \left[2(PR + QX) - V_s^2\right]V_r^2 + (P^2 + Q^2)(R^2 + X^2) = 0$$

input	meaning	condition
`Vs`	sending-end line-to-line voltage (slack)	
`R`	line resistance per phase	
`X`	line reactance per phase	
`P`	load real power (three-phase)	
`Q`	load reactive power in var (negative for a capacitive load)	

output	meaning
`Vr`	receiving-end line voltage
`delta_deg`	receiving-end voltage angle in degrees (negative: lagging the source)
`I`	line current
`P_loss`	real power loss
`Q_loss`	reactive power absorbed by the line in var
`voltage_drop_pct`	(V_s - V_r)/V_s in percent

- **Method:** V_r^4 + [2(PR + QX) - V_s^2] $V_r^2 + (P^2 + Q^2)(R^2 + X^2) = 0$, larger root; angle from $V_s = V_r + (PR + QX)/V_r + j$(PX - QR)/V_r
- **Assumes:** the load is within the line's transfer limit (a real solution exists); the high-voltage (normal) solution is reported

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_two_bus_load_flow', Vs=33000.0, R=0.0, X=10.0, P=0.0, Q=0.0)
# -> Vr = 33000 V, delta_deg = 0, I = 0 A, P_loss = 0 W, Q_loss = 0 var, voltage_drop_pct = 0
(SOLVED)
```

```
energy_systems_ai999x.solve('energy_two_bus_load_flow', Vs=2000.0, R=0.0, X=1.0, P=1000000.0,
Q=0.0)
# -> Vr = 1931.85 V, delta_deg = -15, I = 298.858 A, P_loss = 0 W, Q_loss = 267949 var,
voltage_drop_pct = 3.40742   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Two-bus load flow: Vs = 2 kV, R = 0 ohm, X = 1 ohm, P = 1 MW, Q = 0 var	SOLVED	Vr = 1931.85 V, delta_deg = -15, I = 298.858 A, P_loss = 0 W, Q_loss = 267949 var, voltage_drop_pct = 3.40742
Two-bus load flow: Vs = 33 kV, R = 3 ohm, X = 12 ohm, P = 100 MW, Q = 50 Mvar	INVALID_INPUT	Receiving-end voltage of a line feeding a constant-power load (two-bus load flow): the load exceeds what the line can de

DC power flow — `energy_dc_power_flow`

Bus voltage angles and line flows of a network by the DC (linearised) power flow: lossless lines, flat 1 pu voltages, $P_ij = (theta_i - theta_j)/x_ij$; bus 1 is the slack (angle 0) and takes the balance.

$$\mathbf{B}'\boldsymbol{\theta} = \mathbf{P}, \quad P_{ij} = \frac{\theta_i - \theta_j}{x_{ij}}$$

input	meaning	condition
`lines`	lines as 'i-j x' with the series reactance x in per unit (buses numbered 1..n)	
`P`	net injections (generation minus load) of buses 2..n in per unit	

output	meaning
`theta_deg`	bus voltage angles in degrees, buses 1..n
`flows`	line flows from the first to the second named bus, per unit, in the order given
`P_slack`	slack (bus 1) injection in per unit

- **Method:** B' $\theta = P$ (reduced susceptance matrix, slack removed), solved by Gaussian elimination; $P_ij = (theta_i - theta_j)/x_ij$
- **Assumes:** connected network; positive reactances; bus 1 is the slack
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_dc_power_flow', lines='1-2 0.1, 2-3 0.2', P='0, -1')
# -> theta_deg = [0.0, -5.72957795131, -17.1887338539], flows = [1.0, 1.0], P_slack = 1   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_dc_power_flow', lines='1-2 0.1, 1-2 0.1', P='-1')
# -> theta_deg = [0.0, -2.86478897565], flows = [0.5, 0.5], P_slack = 1   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
DC power flow: lines = 1-2 0.1; 2-3 0.2, P = [0, -1]	SOLVED	theta_deg = [0.0, -5.72957795131, -17.1887338539], flows = [1.0, 1.0], P_slack = 1
DC power flow: lines = 1-2 0.1; 2-3 0.2, P = [0, -1, 2]	INVALID_INPUT	DC power flow: give the net injections of buses 2 to 3 (2 values); bus 1 is the slack.
DC power flow: lines = 1-2 0.1; 2-3 0.2	NEEDS_INPUT	energy_dc_power_flow needs P (net injections (generation minus load) of buses 2..n in per unit). Example: solve('energy_

AC load flow by Newton-Raphson — `energy_newton_load_flow`

Bus voltage magnitudes and angles, slack-bus power, PV-bus reactive power and losses of a small network by the Newton-Raphson load flow (per unit; pi-model lines; bus 1 is the slack, generator buses hold their voltage; reactive limits are not enforced).

$$\begin{bmatrix}\Delta\mathbf{P}\\ \Delta\mathbf{Q}\end{bmatrix} = \begin{bmatrix}\mathbf{H} & \mathbf{N}\\ \mathbf{M} & \mathbf{L}\end{bmatrix}\begin{bmatrix}\Delta\boldsymbol{\theta}\\ \Delta|\mathbf{V}|\end{bmatrix}$$

input	meaning	condition
`lines`	one row per line: from bus, to bus, r, x, total shunt b (per unit)	
`loads`	one row per load: bus, P, Q (per unit)	
`gens`	optional rows per voltage-controlled (PV) generator: bus, P, V (per unit)	
`V1`	slack-bus voltage magnitude in per unit (default 1.0)	

output	meaning
`V`	voltage magnitudes in per unit, buses 1..n
`angle_deg`	voltage angles in degrees, buses 1..n
`P_slack`	slack real power injection (pu)
`Q_slack`	slack reactive power injection (pu)
`P_loss`	total real power loss (pu)
`Q_gen`	reactive power of each PV generator (pu), in the order given

- **Method:** polar Newton-Raphson on the P and Q mismatches with the analytic Jacobian, solved by Gaussian elimination, to a mismatch below $1e-13$ pu
- **Assumes:** buses numbered 1..n, bus 1 the slack; the load flow converges from a flat start
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_newton_load_flow', lines='1 2 0.0 0.5 0.0', loads='2 0.0 0.0')
# -> V = [1.0, 1.0], angle_deg = [0.0, 0.0], P_slack = 0, Q_slack = 0, P_loss = 0, Q_gen = []
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Load flow: lines = [1 2 0 0.5 0], loads = [2 0 0]	SOLVED	V = [1.0, 1.0], angle_deg = [0.0, 0.0], P_slack = 0, Q_slack = 0, P_loss = 0, Q_gen = []
Load flow: lines = [1 2 0.02 0.06 0.03], loads = [2 50 20]	INVALID_INPUT	AC load flow by Newton-Raphson: the load flow diverged (the network cannot carry this loading).
Load flow: loads = [2 0.5 0.2]	NEEDS_INPUT	energy_newton_load_flow needs lines (one row per line: from bus, to bus, r, x, total shunt b (per unit)). Example: solve

Economic dispatch with quadratic costs and limits — `energy_economic_dispatch`

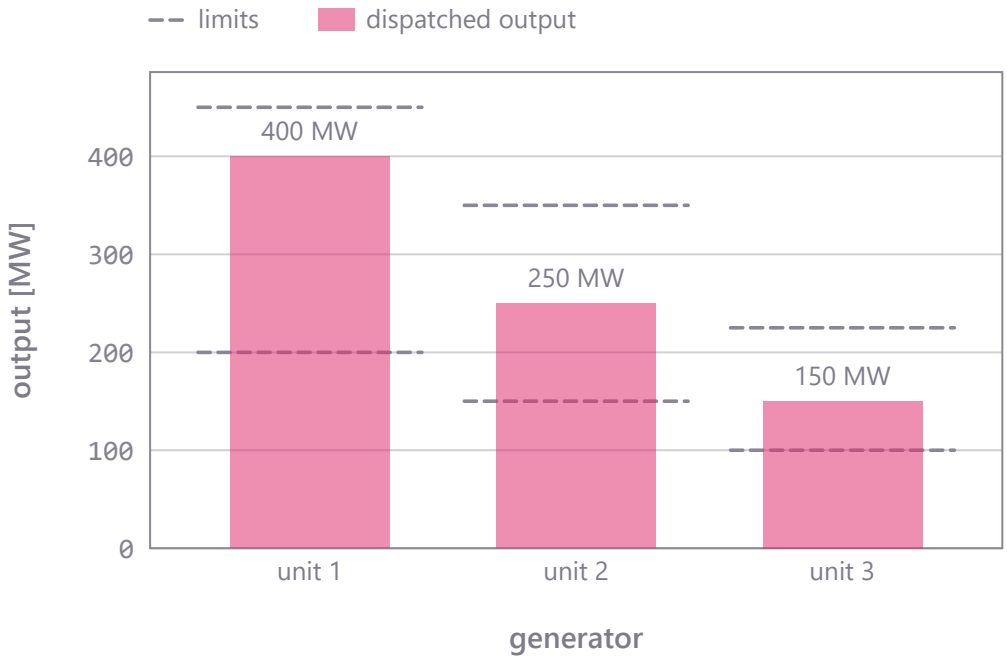
The least-cost outputs of thermal units with costs a + b P + c P^2 ($/h) and output limits that together meet a demand (losses neglected): every unit between its limits runs at the same incremental cost lambda.

$$\frac{dC_i}{dP_i} = b_i + 2c_iP_i = \lambda \ \ (P_{min} < P_i < P_{max}), \quad \sum_i P_i = D$$

input	meaning	condition
`costs`	one row per unit: a ($/h), b($/MWh), c ($/MW^{2}$ h), Pmin (MW), Pmax (MW)	
`demand`	total demand in MW	

output	meaning
`lambda_`	system incremental cost in $/MWh
`P`	unit outputs in MW, in the order given
`cost`	total cost in $/h

- **Method:** equal incremental cost: P_i(lambda) = clip((lambda - b_i)/(2 c_i), Pmin_i, Pmax_i), lambda by bisection on sum $P_i = D$
- **Assumes:** $c > 0$ for every unit; sum Pmin <= demand <= sum Pmax; at least one unit between its limits
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Economic dispatch of 3 generators with quadratic cost curves (rows a, b, c, Pmin, Pmax: 500 5.3 0.004 200 450; 400 5.5 0.006 150 350; 200 5.8 0.009 100 225) meeting a demand of 800 MW: the package returns outputs 400, 250, 150 MW (bars, dashed lines the limits) at a common incremental cost λ = 8.5 per MWh, total cost 6682.5 per hour.

```
energy_systems_ai999x.solve('energy_economic_dispatch', costs='500 5.3 0.004 200 450; 400 5.5 0.006 150 350; 200 5.8 0.009 100 225', demand=800)
# -> lambda_ = 8.5, P = [400.0, 250.0, 150.0], cost = 6682.5   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_economic_dispatch', costs=[[0, 10, 0.01, 0, 500], [0, 10, 0.01, 0, 500]], demand=400)
# -> lambda_ = 14, P = [200.0, 200.0], cost = 4800   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Economic dispatch: costs = [500 5.3 0.004 200 450]; [400 5.5 0.006 150 350]; [200 5.8 0.009 100 225], demand = 800 MW	SOLVED	lambda_ = 8.5, P = [400.0, 250.0, 150.0], cost = 6682.5
Economic dispatch: costs = [500 5.3 0.004 200 450]; [400 5.5 0.006 150 350], demand = 900 MW	INVALID_INPUT	Economic dispatch with quadratic costs and limits: the demand must lie between the total minimum and total maximum outpu

Three-phase (symmetrical) fault current by the bus impedance matrix — `energy_symmetrical_fault`

The fault current, Thevenin reactance and the bus voltages during a balanced three-phase fault at a bus of a reactance network (per unit, prefault no-load voltage Vf everywhere; every branch to node 0 is a source).

$$I_f = \frac{V_f}{Z_{ff} + Z_f}, \quad V_i = V_f - Z_{if} I_f$$

input	meaning	condition
`network`	branches as 'i-j x' with reactances in per unit; node 0 is the reference, each branch to 0 a source (generator or infinite bus)	
`fault_bus`	faulted bus	
`Vf`	prefault voltage in per unit (default 1.0)	
`Xf`	fault reactance in per unit (default 0, a bolted fault)	
`S_base`	base power, for the fault current in amperes (optional)	
`V_base`	base line voltage at the faulted bus (optional)	

output	meaning
`If`	fault current in per unit
`X_th`	Thevenin reactance at the faulted bus (Z_ff) in per unit
`V`	bus voltage magnitudes during the fault (pu), buses 1..n
`If_A`	fault current in amperes

- **Method:** column f of Zbus from Ybus $z = e_f$; $I_f = V_f/(Z_ff + jX_f)$; $V_i = V_f - Z_if$ I_f
- **Assumes:** reactance network; no prefault load current (all prefault voltages Vf)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_symmetrical_fault', network='0-1 0.2, 1-2 0.3', fault_bus=2)
# -> If = 2, X_th = 0.5, V = [0.6, 0.0]   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_symmetrical_fault', network='0-1 0.1', fault_bus=1,
S_base=100000000.0, V_base=100000.0)
# -> If = 10, X_th = 0.1, V = [0.0], If_A = 5773.5 A   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Symmetrical fault: network = 0-1 0.2; 1-2 0.3, fault_bus = 2	SOLVED	If = 2, X_th = 0.5, V = [0.6, 0.0]
Three-phase fault: network = 0-1 0.1, bus = 1, S_base = 100 MVA, V_base = 100 kV	SOLVED	If = 10, X_th = 0.1, V = [0.0], If_A = 5773.5 A
Symmetrical fault: network = 0-1 0.2; 1-2 0.3, fault_bus = 5	INVALID_INPUT	Three-phase (symmetrical) fault current by the bus impedance matrix: the faulted bus must be one of 1..2.

Single line-to-ground fault (symmetrical components) — `energy_slg_fault`

Phase and sequence currents and phase voltages for a fault from phase a to ground through Zf: the three sequence networks in series.

$$I_a = 3I_0 = \frac{3V_f}{Z_1 + Z_2 + Z_0 + 3Z_f}$$

input	meaning	condition
`X1`	positive-sequence Thevenin reactance (pu)	
`X2`	negative-sequence reactance (pu)	
`X0`	zero-sequence reactance (pu)	
`Rf`	fault resistance (pu, default 0)	
`Xf`	fault reactance (pu, default 0)	
`Vf`	prefault voltage (pu, default 1.0)	

output	meaning
`Ia`	|I_a| (pu)
`Ib`	|I_b| (pu)
`Ic`	|I_c| (pu)
`In`	neutral (earth) current |3 I_0| (pu)
`I1`	|I_1| (pu)
`I2`	|I_2| (pu)
`I0`	|I_0| (pu)
`Va`	|V_a| (pu)
`Vb`	|V_b| (pu)
`Vc`	|V_c| (pu)

- **Method:** I1 = I2 = I0 = Vf/(Z1 + Z2 + Z0 + 3 Zf); $\boldsymbol{I_a = 3}$ I_0
- **Assumes:** sequence networks from the Thevenin reactances at the fault; prefault no-load voltage Vf
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_slg_fault', X1=0.1, X2=0.1, X0=0.1)
# -> Ia = 10, Ib = 0, Ic = 0, In = 10, I1 = 3.33333, I2 = 3.33333, I0 = 3.33333, Va = 0, Vb = 1, Vc
= 1   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
SLG fault: X1 = 0.1, X2 = 0.1, X0 = 0.1	SOLVED	Ia = 10, Ib = 0, Ic = 0, In = 10, I1 = 3.33333, I2 = 3.33333, I0 = 3.33333, Va = 0, Vb = 1, Vc = 1
SLG fault: X1 = 0.1, X2 = 0.1, X0 = -0.1	INVALID_INPUT	Single line-to-ground fault (symmetrical components): the sequence reactances must be a positive number.
SLG fault: X1 = 0.1, X2 = 0.1	NEEDS_INPUT	energy_slg_fault needs X0 (zero-sequence reactance (pu)). Example: solve('energy_slg_fault', X1=0.1, X2=0.1, X0=0.1).

Line-to-line fault (symmetrical components) — `energy_ll_fault`

Phase and sequence currents and phase voltages for a fault between phases b and c through Zf: positive and negative networks in parallel opposition.

$$I_b = -I_c = -j\sqrt{3}\,\frac{V_f}{Z_1 + Z_2 + Z_f}$$

input	meaning	condition
`X1`	positive-sequence Thevenin reactance (pu)	
`X2`	negative-sequence reactance (pu)	
`X0`	zero-sequence reactance (pu)	
`Rf`	fault resistance (pu, default 0)	
`Xf`	fault reactance (pu, default 0)	
`Vf`	prefault voltage (pu, default 1.0)	

output	meaning
`Ia`	|I_a| (pu)
`Ib`	|I_b| (pu)
`Ic`	|I_c| (pu)
`In`	neutral (earth) current |3 I_0| (pu)
`I1`	|I_1| (pu)
`I2`	|I_2| (pu)
`I0`	|I_0| (pu)
`Va`	|V_a| (pu)
`Vb`	|V_b| (pu)
`Vc`	|V_c| (pu)

- **Method:** $I_1 = -I_2 = Vf/(Z_1 + Z_2 + Zf)$, $I_0 = 0$; |I_b| = sqrt(3) |I_1|
- **Assumes:** sequence networks from the Thevenin reactances at the fault; prefault no-load voltage Vf
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_ll_fault', X1=0.1, X2=0.1, X0=0.1)
# -> Ia = 0, Ib = 8.66025, Ic = 8.66025, In = 0, I1 = 5, I2 = 5, I0 = 0, Va = 1, Vb = 0.5, Vc = 0.5
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
LL fault: X1 = 0.1, X2 = 0.1, X0 = 0.1	SOLVED	Ia = 0, Ib = 8.66025, Ic = 8.66025, In = 0, I1 = 5, I2 = 5, I0 = 0, Va = 1, Vb = 0.5, Vc = 0.5
LL fault: X1 = 0.1, X2 = 0.1, X0 = -0.1	INVALID_INPUT	Line-to-line fault (symmetrical components): the sequence reactances must be a positive number.
LL fault: X1 = 0.1, X2 = 0.1	NEEDS_INPUT	energy_ll_fault needs X0 (zero-sequence reactance (pu)). Example: solve('energy_ll_fault', X1=0.1, X2=0.1, X0=0.1).

Double line-to-ground fault (symmetrical components) — `energy_dlg_fault`

Phase, neutral and sequence currents and phase voltages for a fault from phases b and c to ground through Zf: negative and (zero + 3Zf) networks in parallel.

$$I_1 = \frac{V_f}{Z_1 + \frac{Z_2(Z_0+3Z_f)}{Z_2+Z_0+3Z_f}}$$

input	meaning	condition
`X1`	positive-sequence Thevenin reactance (pu)	
`X2`	negative-sequence reactance (pu)	
`X0`	zero-sequence reactance (pu)	
`Rf`	fault resistance (pu, default 0)	
`Xf`	fault reactance (pu, default 0)	
`Vf`	prefault voltage (pu, default 1.0)	

output	meaning
`Ia`	|I_a| (pu)
`Ib`	|I_b| (pu)
`Ic`	|I_c| (pu)
`In`	neutral (earth) current |3 I_0| (pu)
`I1`	|I_1| (pu)
`I2`	|I_2| (pu)
`I0`	|I_0| (pu)
`Va`	|V_a| (pu)
`Vb`	|V_b| (pu)
`Vc`	|V_c| (pu)

- **Method:** I1 = Vf/(Z1 + Z2 || $(Z_0 + 3Zf)$); $I_2 = -I_1\ (Z_0 + 3Zf)/(Z_2 + Z_0 + 3Zf)$; $I_0 = -I_1\ Z_2/(Z_2 + Z_0 + 3Zf)$
- **Assumes:** sequence networks from the Thevenin reactances at the fault; prefault no-load voltage Vf
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_dlg_fault', X1=0.1, X2=0.1, X0=0.1)
# -> Ia = 0, Ib = 10, Ic = 10, In = 10, I1 = 6.66667, I2 = 3.33333, I0 = 3.33333, Va = 1, Vb = 0,
Vc = 0   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
DLG fault: X1 = 0.1, X2 = 0.1, X0 = 0.1	SOLVED	Ia = 0, Ib = 10, Ic = 10, In = 10, I1 = 6.66667, I2 = 3.33333, I0 = 3.33333, Va = 1, Vb = 0, Vc = 0
DLG fault: X1 = 0.1, X2 = 0.1, X0 = -0.1	INVALID_INPUT	Double line-to-ground fault (symmetrical components): the sequence reactances must be a positive number.

question	outcome	result
DLG fault: X1 = 0.1, X2 = 0.1	NEEDS_INPUT	energy_dlg_fault needs X0 (zero-sequence reactance (pu)). Example: solve('energy_dlg_fault', X1=0.1, X2=0.1, X0=0.1).

Power transfer and power angle between two voltages — `energy_power_angle`

The power angle needed to transfer P over a lossless reactance X between a sending voltage V2 (generator) and a receiving voltage V1 (infinite bus), the steady-state limit V1 V_2/X, the reactive power at both ends and the stability margin.

$$P = \frac{V_1 V_2}{X}\sin\delta, \quad P_{max} = \frac{V_1 V_2}{X}$$

input	meaning	condition
`V1`	receiving (infinite-bus) line voltage	
`V2`	sending (generator) line voltage	
`X`	transfer reactance per phase	
`P`	real power transferred (three-phase)	

output	meaning
`delta_deg`	power angle (V2 ahead of V1) in degrees
`Pmax`	steady-state limit V1 V_2/X
`Q_sent`	reactive power sent from the generator end in var
`Q_received`	reactive power arriving at the infinite bus in var
`margin_pct`	stability margin (Pmax - P)/Pmax in percent

- **Method:** $P = V_1$ V2 sin(delta)/X; Q_sent = (V2^2 - V1 V2 cos delta)/X; Q_received = (V1 V2 cos delta - V1^2)/X
- **Assumes:** lossless transfer reactance; |P| < V1 V_2/X
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_power_angle', V1=100000.0, V2=100000.0, X=100.0, P=50000000.0)
# -> delta_deg = 30, Pmax = 1e+08 W, Q_sent = 1.33975e+07 var, Q_received = -1.33975e+07 var,
margin_pct = 50   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Power angle: V1 = 100 kV, V2 = 100 kV, X = 100 ohm, P = 50 MW	SOLVED	delta_deg = 30, Pmax = 1e+08 W, Q_sent = 1.33975e+07 var, Q_received = -1.33975e+07 var, margin_pct = 50
Power angle: V1 = 100 kV, V2 = 100 kV, X = 100 ohm, P = 150 MW	INVALID_INPUT	Power transfer and power angle between two voltages: the power exceeds the steady-state limit V1 V2/X.

Critical clearing angle and time (equal-area criterion) — `energy_critical_clearing`

The initial rotor angle, the largest allowable angle and the critical clearing angle of a machine against an infinite bus by the equal-area criterion, for given pre-fault, during-fault and post-fault transfer limits; with H and f, the critical clearing time for a fault that blocks all transfer.

$$\cos\delta_{cr} = \frac{P_m(\delta_{max} - \delta_0) + P_{max,3}\cos\delta_{max} - P_{max,2}\cos\delta_0}{P_{max,3} - P_{max,2}}$$

input	meaning	condition
`Pm`	mechanical input power (pu)	
`Pmax_pre`	pre-fault transfer limit (pu)	
`Pmax_fault`	transfer limit during the fault (pu; 0 for a fault at the terminals)	
`Pmax_post`	post-fault transfer limit (pu)	
`H`	inertia constant H (MJ/MVA, i.e. seconds), for the clearing time	
`f`	system frequency, for the clearing time	

output	meaning
`delta0_deg`	initial rotor angle in degrees
`delta_max_deg`	largest angle before loss of synchronism in degrees
`delta_cr_deg`	critical clearing angle in degrees
`t_cr`	critical clearing time (Pmax_fault = 0)

- **Method:** cos(delta_cr) = [Pm (delta_max - delta0) + Pmax_post cos(delta_max) - Pmax_fault cos(delta0)]/(Pmax_post - Pmax_fault); $t_cr = sqrt$(4 H (delta_cr - delta0)/(omega_s Pm))
- **Assumes:** Pmax_fault < Pm < Pmax_pre, Pmax_post; classical model (constant voltage behind transient reactance, no damping)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_critical_clearing', Pm=1.0, Pmax_pre=2.0, Pmax_fault=0.0,
Pmax_post=2.0)
# -> delta0_deg = 30, delta_max_deg = 150, delta_cr_deg = 79.562   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Critical clearing: Pm = 1, Pmax_pre = 2, Pmax_fault = 0, Pmax_post = 2	SOLVED	delta0_deg = 30, delta_max_deg = 150, delta_cr_deg = 79.562
Critical clearing: Pm = 1, Pmax_pre = 2, Pmax_fault = 1.5, Pmax_post = 2	INVALID_INPUT	Critical clearing angle and time (equal-area criterion): the during-fault transfer limit must be below the mechanical po

Frequency deviation and load sharing under governor droop — `energy_droop_sharing`

The steady-state frequency drop after a load step and each generator's output change when governors with droop R (per unit on their own rating) share the load, with optional load damping D.

$$\Delta f = -\frac{\Delta P_L}{\sum_i \frac{S_i}{R_i f_0} + D}$$

input	meaning	condition
`R`	droop of each governor in per unit on its own rating (e.g. 0.05)	
`ratings`	rating of each generator in MW	
`dP`	load increase in MW (negative for a load decrease)	
`f0`	nominal frequency	
`D`	load damping in MW/Hz (default 0)	

output	meaning
`df`	frequency change in Hz
`f`	new frequency in Hz
`dP_gen`	output change of each generator in MW
`dP_load_damping`	load relief from damping in MW

- **Method:** beta = sum(S_i/(R_i f0)) + D; $df = -dP/\beta$; $dP_i = -df$ S_i/(R_i f0)
- **Assumes:** primary control only (no secondary/AGC); R, ratings > 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_droop_sharing', R=[0.05, 0.05], ratings=[500, 500], dP=100,
f0=50)
# -> df = -0.25, f = 49.75, dP_gen = [50.0, 50.0], dP_load_damping = 0   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Droop sharing: R = [0.05, 0.05], ratings = [500, 500], dP = 100 MW, f0 = 50 Hz	SOLVED	df = -0.25, f = 49.75, dP_gen = [50.0, 50.0], dP_load_damping = 0
Droop sharing: R = [0.05, 0.05], ratings = [500], dP = 100 MW, f0 = 50 Hz	INVALID_INPUT	Frequency deviation and load sharing under governor droop: give one droop and one rating per generator.

Transformer regulation and efficiency from its equivalent circuit — `energy_transformer_performance`

Primary voltage needed, voltage regulation, primary current and input power factor, copper and core losses and efficiency of a single-phase transformer (approximate equivalent circuit referred to the primary, shunt branch at the primary terminals) delivering a fraction of its rating at rated secondary voltage.

$$\mathbf{V}_1 = a\mathbf{V}_2 + \frac{\mathbf{I}_2}{a}(R_{eq} + jX_{eq}), \quad \eta = \frac{P_{out}}{P_{out} + P_{cu} + P_{core}}$$

input	meaning	condition
`S`	rating in VA	

input	meaning	condition
`V1`	rated primary voltage	
`V2`	rated secondary voltage	
`Req`	series resistance referred to the primary	
`Xeq`	series leakage reactance referred to the primary	
`Rc`	core-loss resistance (primary side)	
`Xm`	magnetizing reactance (primary side)	
`load`	load as a fraction of the rating (or a percentage)	
`pf`	load power factor	
`pf_type`	lagging (default) or leading load	

output	meaning
`V1_required`	primary voltage for rated secondary voltage at this load
`regulation_pct`	voltage regulation in percent
`I1`	primary current
`pf_in`	input power factor
`P_out`	output power
`P_cu`	copper loss
`P_core`	core loss
`efficiency_pct`	efficiency in percent

- **Method:** $V_1 = a\,V_2 + I_2$' (Req + jXeq), I2' = I_2/a at angle -acos(pf); P_cu = |I2'|^2 Req; P_core = V1^2/Rc; eta = P_out/(P_out + P_cu + P_core)
- **Assumes:** approximate equivalent circuit (shunt branch at the primary terminals); 0 < load <= 2
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_transformer_performance', S=10000.0, V1=1000, V2=100, Req=1.0,
Xeq=0.0, Rc=1000000.0, Xm=1000000000.0, load=1.0, pf=1.0)
# -> V1_required = 1010 V, regulation_pct = 1, I1 = 10.001 A, pf_in = 1, P_out = 10000 W, P_cu =
100 W, P_core = 1.0201 W, efficiency_pct = 98.9999   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_transformer_performance', S=10000.0, V1=1000, V2=100, Req=1.0,
Xeq=0.0, Rc=1000000000.0, Xm=1000000000.0, load=1.0, pf=1.0)
# -> V1_required = 1010 V, regulation_pct = 1, I1 = 10 A, pf_in = 1, P_out = 10000 W, P_cu = 100 W,
P_core = 0.0010201 W, efficiency_pct = 99.0099   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Transformer regulation and efficiency: S = 10 kVA, V1 = 1000 V, V2 = 100 V, Req = 1 ohm, Xeq = 0 ohm, Rc = 1e9 ohm, Xm = 1e9 ohm, load = 100%, pf = 1	SOLVED	V1_required = 1010 V, regulation_pct = 1, I1 = 10 A, pf_in = 1, P_out = 10000 W, P_cu = 100 W, P_core = 0.0010201 W, efficiency_pct = 99.0099
Transformer regulation and efficiency: S = 10 kVA, V1 = 1000 V, V2 = 100 V, Req = 1 ohm, Xeq = 0 ohm, Rc = 1e9 ohm, Xm = 1e9 ohm, load = 300%, pf = 1	INVALID_INPUT	Transformer regulation and efficiency from its equivalent circuit: the load must lie between 0 and 2 times the rating (a

Transformer parameters from open-circuit and short-circuit tests — `energy_transformer_tests`

Core-loss resistance and magnetizing reactance from the open-circuit test, and the equivalent series resistance, reactance and impedance from the short-circuit test (each on the side where the test was made).

$$R_c = \frac{V_{oc}^2}{P_{oc}},\ X_m = \frac{1}{\sqrt{Y_{oc}^2 - G_c^2}},\ R_{eq} = \frac{P_{sc}}{I_{sc}^2},\ X_{eq} = \sqrt{Z_{eq}^2 - R_{eq}^2}$$

input	meaning	condition
`Voc`	open-circuit test voltage	
`Ioc`	open-circuit test current	
`Poc`	open-circuit test power	
`Vsc`	short-circuit test voltage	
`Isc`	short-circuit test current	
`Psc`	short-circuit test power	

output	meaning
`Rc`	core-loss resistance (open-circuit side)
`Xm`	magnetizing reactance (open-circuit side)
`Req`	equivalent series resistance (short-circuit side)
`Xeq`	equivalent leakage reactance (short-circuit side)
`Zeq`	equivalent series impedance magnitude
`pf_oc`	open-circuit power factor
`pf_sc`	short-circuit power factor

- **Method:** $Rc = V_oc^2/P_oc$, $Xm = \frac{1}{\sqrt{\left(\frac{I_{oc}}{V_{oc}}\right)^2 - \left(\frac{P_{oc}}{V_{oc}^2}\right)^2}}$; Req = P_sc/I_sc^2, $Xeq = \sqrt{\left(\frac{V_{sc}}{I_{sc}}\right)^2 - Req^2}$
- **Assumes:** $P < V$ I in both tests
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_transformer_tests', Voc=100, Ioc=1, Poc=60, Vsc=50, Isc=10,
Psc=300)
# -> Rc = 166.667 ohm, Xm = 125 ohm, Req = 3 ohm, Xeq = 4 ohm, Zeq = 5 ohm, pf_oc = 0.6, pf_sc =
0.6   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Transformer tests: Voc = 100 V, Ioc = 1 A, Poc = 60 W, Vsc = 50 V, Isc = 10 A, Psc = 300 W	SOLVED	Rc = 166.667 ohm, Xm = 125 ohm, Req = 3 ohm, Xeq = 4 ohm, Zeq = 5 ohm, pf_oc = 0.6, pf_sc = 0.6
Transformer tests: Voc = 100 V, Ioc = 1 A, Poc = 160 W, Vsc = 50 V, Isc = 10 A, Psc = 300 W	INVALID_INPUT	Transformer parameters from open-circuit and short-circuit tests: the open-circuit power must be below V I (power factor

Load for maximum transformer efficiency — `energy_transformer_max_efficiency`

The load (fraction of rating and apparent power) at which a transformer's efficiency is greatest (copper loss equal to core loss), that maximum efficiency and the full-load efficiency at a power factor.

$$x_{\eta,max} = \sqrt{\frac{P_{core}}{P_{cu,fl}}}, \quad \eta_{max} = \frac{x S \cos\phi}{x S \cos\phi + 2P_{core}}$$

input	meaning	condition
`S`	rating in VA	
`P_core`	core (no-load) loss	
`P_cu_fl`	copper loss at full load	
`pf`	load power factor	

output	meaning
`load_fraction`	load fraction for maximum efficiency sqrt(P_core/P_cu)
`S_at_max`	apparent power at maximum efficiency in VA
`efficiency_max_pct`	maximum efficiency in percent
`efficiency_full_load_pct`	full-load efficiency in percent

- **Method:** $x = sqrt$(P_core/P_cu,fl); eta_max = x S pf/(x S $pf + 2$ P_core)
- **Assumes:** constant core loss, copper loss proportional to the load squared
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_transformer_max_efficiency', S=100000.0, P_core=500,
P_cu_fl=2000, pf=1.0)
# -> load_fraction = 0.5, S_at_max = 50000 VA, efficiency_max_pct = 98.0392,
efficiency_full_load_pct = 97.561   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Maximum transformer efficiency: S = 100 kVA, P_core = 500 W, P_cu_fl = 2000 W, pf = 1	SOLVED	load_fraction = 0.5, S_at_max = 50000 VA, efficiency_max_pct = 98.0392, efficiency_full_load_pct = 97.561
Maximum transformer efficiency: S = 100 kVA, P_core = 0 W, P_cu_fl = 2000 W, pf = 1	INVALID_INPUT	Load for maximum transformer efficiency: the rating and the losses must be a positive number.

Transformer under load from nameplate data — `energy_transformer_loading`

Secondary voltage, losses, efficiency and loading of a transformer fed at rated primary voltage and supplying a P + jQ load, from its nameplate: rating, rated voltages, short-circuit voltage vk and its resistive part vkr, no-load loss and no-load current (T equivalent circuit).

$$z_k = v_{kr} + j\sqrt{v_k^2 - v_{kr}^2}, \quad y_m = \frac{P_0}{S_n} - j\sqrt{i_0^2 - \left(\frac{P_0}{S_n}\right)^2}$$

input	meaning	condition
`sn`	rating in VA	
`V_hv`	rated (and applied) high-voltage side voltage	
`V_lv`	rated low-voltage side voltage	
`vk`	short-circuit voltage as a fraction or percentage of rated voltage	
`vkr`	resistive part of the short-circuit voltage (fraction or %)	
`P0`	no-load (iron) loss	
`i0`	no-load current as a fraction or percentage of rated current	
`P`	load real power on the low-voltage side	
`Q`	load reactive power in var	

output	meaning
`V_lv_actual`	low-voltage side voltage under load
`V_lv_pu`	low-voltage side voltage in per unit
`delta_deg`	phase shift of the secondary voltage in degrees (no vector-group shift)
`P_loss`	total real loss
`Q_loss`	reactive power absorbed in var
`efficiency_pct`	efficiency in percent
`loading_pct`	loading: the larger winding current over its rated current, percent

- **Method:** per unit on the rating: z = vkr + j sqrt(vk^2 - vkr^2), $y_m = P_0/S - j$ sqrt(i0^2 - (P0/S)^2); the T circuit solved for the load-bus voltage by Newton iteration
- **Assumes:** T equivalent circuit (series impedance halved on each side of the magnetizing branch); rated voltage applied, nominal tap
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_transformer_loading', sn=630000.0, V_hv=20000.0, V_lv=400,
vk=0.06, vkr=0.012, P0=1500, i0=0.003, P=400000.0, Q=150000.0)
# -> V_lv_actual = 390.876 V, V_lv_pu = 0.977189, delta_deg = -2.02483, P_loss = 5120.79 W, Q_loss
= 19030.9 var, efficiency_pct = 98.736, loading_pct = 69.6777   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Transformer under load: sn = 630 kVA, V_hv = 20 kV, V_lv = 400 V, vk = 6%, vkr = 1.2%, P0 = 1.5 kW, i0 = 0.3%, P = 400 kW, Q = 150 kvar	SOLVED	V_lv_actual = 390.876 V, V_lv_pu = 0.977189, delta_deg = -2.02483, P_loss = 5120.79 W, Q_loss = 19030.9 var, efficiency_pct = 98.736, loading_pct = 69.6777
Transformer under load: sn = 630 kVA, V_hv = 20 kV, V_lv = 400 V, vk = 6%, vkr = 7%, P0 = 1.5 kW, i0 = 0.3%, P = 400 kW, Q = 150 kvar	INVALID_INPUT	Transformer under load from nameplate data: need 0 < vkr < vk (short-circuit voltages as fractions or percentages, vk be

Load sharing between two transformers in parallel — `energy_parallel_transformers`

How two transformers with equal voltage ratios but different ratings and short-circuit impedances share a load: the apparent, real and reactive power each delivers and its loading on its own rating.

$$\mathbf{S}_1 = \mathbf{S}_L\left(\frac{\mathbf{Z}_2}{\mathbf{Z}_1+\mathbf{Z}_2}\right)^*, \quad \mathbf{S}_2 = \mathbf{S}_L\left(\frac{\mathbf{Z}_1}{\mathbf{Z}_1+\mathbf{Z}_2}\right)^*$$

input	meaning	condition
`S1`	rating of transformer 1 (VA)	
`vk1`	short-circuit voltage of transformer 1 (fraction or %)	
`vkr1`	its resistive part (fraction or %)	
`S2`	rating of transformer 2 (VA)	
`vk2`	short-circuit voltage of transformer 2 (fraction or %)	
`vkr2`	its resistive part (fraction or %)	
`V_lv`	common secondary rated voltage	
`P`	load real power	
`Q`	load reactive power in var	

output	meaning
`S1_share`	apparent power delivered by transformer 1 in VA
`S2_share`	apparent power delivered by transformer 2 in VA
`P1`	real power of transformer 1
`Q1`	reactive power of transformer 1 in var
`P2`	real power of transformer 2
`Q2`	reactive power of transformer 2 in var
`share1_pct_of_rating`	transformer 1 output over its rating, percent
`share2_pct_of_rating`	transformer 2 output over its rating, percent

- **Method:** impedances in ohms on the common secondary: Z_i = (vkr_i + j sqrt(vk_i^2 - vkr_i^2)) V^2/S_i; $S_1 = S_L$ conj($Z_2/(Z_1+Z_2)$), $S_2 = S_L$ conj($Z_1/(Z_1+Z_2)$)
- **Assumes:** equal voltage ratios (no circulating current); magnetizing branches neglected
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_parallel_transformers', S1=500000.0, vk1=0.05, vkr1=0.01,
S2=500000.0, vk2=0.05, vkr2=0.01, V_lv=400, P=600000.0, Q=200000.0)
# -> S1_share = 316228 VA, S2_share = 316228 VA, P1 = 300000 W, Q1 = 100000 var, P2 = 300000 W, Q2
= 100000 var, share1_pct_of_rating = 63.2456, share2_pct_of_rating = 63.2456   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Parallel transformers: S1 = 500 kVA, vk1 = 5%, vkr1 = 1%, S2 = 500 kVA, vk2 = 5%,	SOLVED	S1_share = 316228 VA, S2_share = 316228 VA, P1 = 300000 W, Q1 = 100000 var, P2 = 300000 W, Q2 = 100000 var,

question	outcome	result
vkr2 = 1%, V_lv = 400 V, P = 600 kW, Q = 200 kvar		share1_pct_of_rating = 63.2456, share2_pct_of_rating = 63.2456
Parallel transformers: S1 = 500 kVA, vk1 = 5%, vkr1 = 6%, S2 = 500 kVA, vk2 = 5%, vkr2 = 1%, V_lv = 400 V, P = 600 kW, Q = 200 kvar	INVALID_INPUT	Load sharing between two transformers in parallel: need 0 <= vkr < vk below 50% for each transformer.

Induction motor performance from its equivalent circuit — `energy_induction_motor`

Line and rotor currents, power factor, air-gap and converted power, induced torque, output and efficiency of a three-phase induction motor at a slip, from its per-phase equivalent circuit.

$$T_{ind} = \frac{3 I_2^2 R_2 / s}{\omega_s}, \quad n_s = \frac{120 f}{p}$$

input	meaning	condition
`VL`	line voltage	
`f`	supply frequency	
`poles`	number of poles (even)	
`R1`	stator resistance per phase	
`X1`	stator leakage reactance per phase	
`R2`	rotor resistance per phase referred to the stator	
`X2`	rotor leakage reactance referred to the stator	
`Xm`	magnetizing reactance per phase	
`slip`	slip $(0 < s \le 1)$	
`connection`	stator connection	
`P_rot`	rotational (friction, windage and core) loss, default 0	

output	meaning
`I_line`	line current
`I2`	rotor current referred to the stator
`pf`	power factor
`P_in`	input power
`P_ag`	air-gap power
`P_conv`	converted (developed) power
`T`	induced torque
`P_out`	output power
`efficiency_pct`	efficiency in percent

- **Method:** Z = R1 + jX1 + jXm || $(R_2/s + jX2)$; $P_ag = 3\ I_2^2\ R_2/s$; P_conv = (1 - s) P_ag; $T = P_ag/omega_s$; $n_s = 120$ f/p
- **Assumes:** steady state, balanced supply; 0 < slip <= 1

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_induction_motor', VL=692.8203230275509, f=50, poles=4, R1=0.0,
X1=2.0, R2=1.5, X2=2.0, Xm=1000000000000.0, slip=0.5, connection='star')
# -> I_line = 80 A, I2 = 80 A, pf = 0.6, P_in = 57600 W, P_ag = 57600 W, P_conv = 28800 W, T =
366.693 N*m, P_out = 28800 W, efficiency_pct = 50   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Induction motor: VL = 460 V, f = 60 Hz, poles = 4, R1 = 0.641 ohm, X1 = 1.106 ohm, R2 = 0.332 ohm, X2 = 0.464 ohm, Xm = 26.3 ohm, slip = 2.2%, connection = star	SOLVED	I_line = 18.8919 A, I2 = 16.171 A, pf = 0.832122, P_in = 12525.1 W, P_ag = 11838.8 W, P_conv = 11578.4 W, T = 62.8068 N*m, P_out = 11578.4 W, efficiency_pct = 92.4409
Induction motor: VL = 460 V, f = 60 Hz, poles = 5, R1 = 0.641 ohm, X1 = 1.106 ohm, R2 = 0.332 ohm, X2 = 0.464 ohm, Xm = 26.3 ohm, slip = 2.2%, connection = star	INVALID_INPUT	Induction motor performance from its equivalent circuit: the number of poles must be an even whole number (2, 4, 6, ...)

Induction motor breakdown (maximum) torque and starting torque —

`energy_induction_max_torque`

The slip at maximum torque, the maximum (breakdown) torque, the starting torque and current, and the Thevenin equivalent of the stator and magnetizing branch, for a three-phase induction motor.

$$s_{max} = \frac{R_2}{\sqrt{R_{th}^2 + (X_{th} + X_2)^2}}, \quad T_{max} = \frac{3V_{th}^2}{2\omega_s \left[R_{th} + \sqrt{R_{th}^2 + (X_{th} + X_2)^2}\right]}$$

input	meaning	condition
`VL`	line voltage	
`f`	supply frequency	
`poles`	number of poles (even)	
`R1`	stator resistance per phase	
`X1`	stator leakage reactance per phase	
`R2`	rotor resistance per phase referred to the stator	
`X2`	rotor leakage reactance referred to the stator	
`Xm`	magnetizing reactance per phase	
`connection`	stator connection	

output	meaning
`s_maxT`	slip at maximum torque
`T_max`	maximum (pull-out) torque
`T_start`	starting torque $(s = 1)$
`I_start`	starting line current

output	meaning
`V_th`	Thevenin voltage per phase
`R_th`	Thevenin resistance
`X_th`	Thevenin reactance

- **Method:** Thevenin: $V_th = V$ jXm/(R1 + j(X1 + Xm)), $Z_th = Z_1$ || jXm; $s_{max} = \frac{R_2}{\sqrt{R_{th}^2+(X_{th}+X_2)^2}}$; T_max = 3 V_th^2/(2 omega_s (R_th + sqrt(R_th^2 + (X_th + X2)^2)))
- **Assumes:** constant parameters (no deep-bar or saturation effects)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_induction_max_torque', VL=692.8203230275509, f=50, poles=4,
R1=0.0, X1=1.0, R2=0.2, X2=1.0, Xm=1000000000000.0, connection='star')
# -> s_maxT = 0.1, T_max = 763.944 N*m, T_start = 151.276 N*m, I_start = 199.007 A, V_th = 400 V,
R_th = 0 ohm, X_th = 1 ohm   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Induction motor maximum torque: VL = 460 V, f = 60 Hz, poles = 4, R1 = 0.641 ohm, X1 = 1.106 ohm, R2 = 0.332 ohm, X2 = 0.464 ohm, Xm = 26.3 ohm	NEEDS_INPUT	energy_induction_max_torque needs connection (stator connection). Example: solve('energy_induction_max_torque', VL='692.
Induction motor breakdown torque: VL = 460 V, f = 60 Hz, poles = 4, R1 = 0 ohm, X1 = 1 ohm, R2 = 0.2 ohm, X2 = 1 ohm, Xm = 1e12 ohm, connection = star	SOLVED	s_maxT = 0.1, T_max = 280.643 N*m, T_start = 55.5729 N*m, I_start = 132.132 A, V_th = 265.581 V, R_th = 0 ohm, X_th = 1 ohm

Synchronous generator internal EMF, power angle and regulation —

`energy_synchronous_generator`

The internal (excitation) EMF, power angle and voltage regulation of a star-connected cylindrical-rotor synchronous generator delivering a current at a power factor, from its synchronous reactance and armature resistance.

$$\mathbf{E}_A = \mathbf{V}_\phi + (R_A + jX_S)\mathbf{I}_A$$

input	meaning	condition
`VL`	terminal line voltage	
`IL`	armature (line) current	
`pf`	power factor	
`Xs`	synchronous reactance per phase	
`Ra`	armature resistance per phase (default 0)	
`pf_type`	lagging (default) or leading	

output	meaning
`E_line`	internal EMF (line value)
`E_phase`	internal EMF per phase
`delta_deg`	power (torque) angle in degrees
`regulation_pct`	voltage regulation (E - V)/V in percent
`P`	real power output
`Q`	reactive power output in var

- **Method:** E = V + (Ra + jXs) I per phase, I at angle -acos(pf)
- **Assumes:** cylindrical rotor (no saliency), star connection, unsaturated
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_synchronous_generator', VL=173.20508075688772, IL=10, pf=1.0, Xs=10.0)
# -> E_line = 244.949 V, E_phase = 141.421 V, delta_deg = 45, regulation_pct = 41.4214, P = 3000 W, Q = 0 var   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Synchronous generator: VL = 173.20508075688772 V, IL = 10 A, pf = 1, Xs = 10 ohm	SOLVED	E_line = 244.949 V, E_phase = 141.421 V, delta_deg = 45, regulation_pct = 41.4214, P = 3000 W, Q = 0 var
Synchronous generator: VL = 480 V, IL = 60 A, pf = 0.8, Xs = 0 ohm	INVALID_INPUT	Synchronous generator internal EMF, power angle and regulation: the terminal voltage, the current and the synchronous re

Peukert exponent and run time at a new discharge current — `energy_peukert`

Peukert's law $t = C_p / I^k$ fitted (least squares on the logarithms) to two or more constant-current discharge tests, then the run time and delivered capacity at another current.

$$t = \frac{C_p}{I^k} \;\Rightarrow\; \ln t = \ln C_p - k \ln I$$

input	meaning	condition
`currents`	test discharge currents in A	
`times`	run time of each test in hours	
`I`	new discharge current	

output	meaning
`k`	Peukert exponent
`C_p`	Peukert capacity C_p in A^k h
`runtime_h`	run time at I in hours
`capacity_Ah`	capacity delivered at I in Ah

- **Method:** ln $t = ln$ C_p - k ln I fitted by least squares; t(I) = C_p I^{-k}
- **Assumes:** at least two tests at different currents; constant temperature and cut-off voltage
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_peukert', currents=[1, 10], times=[100, 10], I=5)
# -> k = 1, C_p = 100, runtime_h = 20, capacity_Ah = 100   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_peukert', currents='2, 8', times='50, 10', I=4)
# -> k = 1.16096, C_p = 111.803, runtime_h = 22.3607, capacity_Ah = 89.4427   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Peukert: currents = [1, 10], times = [100, 10], I = 5 A	SOLVED	k = 1, C_p = 100, runtime_h = 20, capacity_Ah = 100
Peukert: currents = [5, 5], times = [20, 4], I = 10 A	INVALID_INPUT	Peukert exponent and run time at a new discharge current: the test currents must be different.

Battery pack series-parallel sizing — `energy_battery_pack_sizing`

The number of cells in series to reach a pack voltage and of strings in parallel to reach a pack energy, with the resulting pack voltage, energy and capacity (fewest cells; nominal values).

$$N_s = \left\lceil \frac{V_{pack}}{V_{cell}} \right\rceil, \quad N_p = \left\lceil \frac{E_{pack}}{N_s V_{cell} Q_{cell}} \right\rceil$$

input	meaning	condition
`V_target`	required nominal pack voltage	
`E_target`	required pack energy (Wh or kWh)	
`V_cell`	nominal cell voltage	
`Q_cell`	cell capacity (Ah accepted)	

output	meaning
`N_series`	cells in series
`N_parallel`	strings in parallel
`N_cells`	total cells
`V_pack`	nominal pack voltage
`E_pack_kWh`	pack energy in kWh
`Q_pack_Ah`	pack capacity in Ah

- **Method:** N_s = ceil(V/V_cell); N_p = ceil(E/(N_s V_cell Q_cell))
- **Assumes:** nominal voltages and capacities; a target met within $1e-9$ counts as met
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_battery_pack_sizing', V_target=48, E_target=36000000.0,
V_cell=3.2, Q_cell=360000)
# -> N_series = 15, N_parallel = 3, N_cells = 45, V_pack = 48 V, E_pack_kWh = 14.4, Q_pack_Ah = 300
(SOLVED)
```

```
energy_systems_ai999x.solve('energy_battery_pack_sizing', V_target=400, E_target=216000000.0,
V_cell=3.7, Q_cell=18000)
# -> N_series = 109, N_parallel = 30, N_cells = 3270, V_pack = 403.3 V, E_pack_kWh = 60.495,
Q_pack_Ah = 150   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Battery pack sizing: V_target = 48 V, E_target = 10 kWh, V_cell = 3.2 V, Q_cell = 100 Ah	SOLVED	N_series = 15, N_parallel = 3, N_cells = 45, V_pack = 48 V, E_pack_kWh = 14.4, Q_pack_Ah = 300
Battery pack sizing: V_target = 48 V, E_target = 10 kWh, V_cell = 0 V, Q_cell = 100 Ah	INVALID_INPUT	Battery pack series-parallel sizing: the target voltage and energy and the cell voltage and capacity must be a positive

Battery terminal voltage under constant current (one-RC equivalent circuit) — `energy_battery_rc_discharge`

Terminal voltage, polarization (RC) voltage, energy delivered and heat generated after a constant-current discharge of duration t, for a cell modelled as an open-circuit voltage with a series resistance R0 and one R1 || C1 branch (constant OCV).

$$V_t(t) = V_{oc} - IR_0 - IR_1\left(1 - e^{-t/R_1C_1}\right)$$

input	meaning	condition
`OCV`	open-circuit voltage (held constant)	
`R0`	series (ohmic) resistance	
`R1`	polarization resistance	
`C1`	polarization capacitance	
`I`	discharge current	
`t`	time since the current step	

output	meaning
`V_terminal`	terminal voltage
`V_rc`	voltage across the RC branch
`E_delivered`	energy delivered to the load
`E_Wh`	energy delivered in Wh
`heat`	heat generated in R0 and R1
`tau`	time constant R1 C1

- **Method:** $V_1 = I$ R1 (1 - e^(-t/tau)), V_t = OCV - I $R_0 - V_1$; energy and heat integrated in closed form
- **Assumes:** constant open-circuit voltage over the interval (short pulse or flat OCV curve); RC branch initially relaxed

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_battery_rc_discharge', OCV=4.0, R0=0.01, R1=0.02, C1=1000,
I=10, t=0)
# -> V_terminal = 3.9 V, V_rc = 0 V, E_delivered = 0 J, E_Wh = 0, heat = 0 J, tau = 20 s   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_battery_rc_discharge', OCV=4.0, R0=0.01, R1=0.02, C1=1000,
I=10, t=13.862943611198906)
# -> V_terminal = 3.8 V, V_rc = 0.1 V, E_delivered = 532.929 J, E_Wh = 0.148036, heat = 16.5888 J,
tau = 20 s   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Battery RC model: OCV = 4 V, R0 = 10 mohm, R1 = 20 mohm, C1 = 1000 F, I = 10 A, t = 0 s	SOLVED	V_terminal = 3.9 V, V_rc = 0 V, E_delivered = 0 J, E_Wh = 0, heat = 0 J, tau = 20 s
Battery RC model: OCV = 4 V, R0 = 10 mohm, R1 = 20 mohm, C1 = 1000 F, I = 500 A, t = 60 s	INVALID_INPUT	Battery terminal voltage under constant current (one-RC equivalent circuit): the terminal voltage would fall to zero: th

Hydrogen fuel-cell stack: consumption, power and efficiency — `energy_fuel_cell_stack`

Hydrogen and oxygen consumption, water production (Faraday's law), efficiency on the lower and higher heating value and heat released of a hydrogen fuel-cell stack of N cells at a current and cell voltage.

$$\dot{n}_{H_2} = \frac{NI}{2F}, \quad \eta_{LHV} = \frac{NIV_{cell}}{\dot{n}_{H_2}\,\Delta h_{LHV}}$$

input	meaning	condition
`I`	stack current	
`N`	number of cells in series	
`V_cell`	average cell voltage	
`utilization`	fuel utilization (fraction or %, default 1)	

output	meaning
`H2_kg_s`	hydrogen consumed
`H2_g_h`	hydrogen consumed in g/h
`H2_fed_kg_s`	hydrogen fed (consumed/utilization)
`O2_kg_s`	oxygen consumed
`H2O_kg_s`	water produced
`efficiency_LHV`	efficiency on the lower heating value
`efficiency_HHV`	efficiency on the higher heating value
`heat_LHV`	heat released (water vapour product)

- **Method:** $n_H2 = N\,I/(2F)$; $n_O2 = n_H2/2$; $n_H2O = n_H2$; $P = N$ I V; eta = P/(n_H2 LHV)
- **Assumes:** hydrogen/oxygen cell, 2 electrons per H2; heating values at 25 degC

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_fuel_cell_stack', I=192970.66424, N=1, V_cell=0.7)
# -> H2_kg_s = 0.00201588 kg/s, H2_g_h = 7257.17, H2_fed_kg_s = 0.00201588 kg/s, O2_kg_s =
0.0159994 kg/s, H2O_kg_s = 0.0180153 kg/s, efficiency_LHV = 0.558581, efficiency_HHV = 0.472593,
heat_LHV = 106747 W   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Fuel-cell stack: I = 192970.66424 A, N = 1, V_cell = 0.7 V	SOLVED	H2_kg_s = 0.00201588 kg/s, H2_g_h = 7257.17, H2_fed_kg_s = 0.00201588 kg/s, O2_kg_s = 0.0159994 kg/s, H2O_kg_s = 0.0180153 kg/s, efficiency_LHV = 0.558581, efficiency_HHV = 0.472593, heat_LHV = 106747 W
Fuel-cell stack: I = 200 A, N = 100, V_cell = 1.4 V	INVALID_INPUT	Hydrogen fuel-cell stack: consumption, power and efficiency: a hydrogen fuel cell's voltage must stay below the thermone

Water electrolyser: hydrogen production and energy per kilogram — `energy_electrolyser`

Hydrogen and oxygen production, water consumption, specific energy (kWh per kg H2) and efficiency on the higher and lower heating value of a water electrolyser of N cells.

$$\frac{E}{m_{H_2}} = \frac{2FV_{cell}}{\eta_F M_{H_2}}$$

input	meaning	condition
`I`	stack current	
`N`	number of cells in series	
`V_cell`	cell voltage (1.23 to 3 V)	
`faradaic`	Faradaic (current) efficiency, fraction or % (default 1)	

output	meaning
`H2_kg_s`	hydrogen produced
`H2_kg_h`	hydrogen produced in kg/h
`O2_kg_s`	oxygen produced
`H2O_kg_s`	water consumed
`specific_energy_kWh_per_kg`	electricity per kg of hydrogen in kWh/kg
`efficiency_HHV`	efficiency on the higher heating value
`efficiency_LHV`	efficiency on the lower heating value

- **Method:** $n_H2 = eta_F$ N $I/(2F)$; specific energy = N I V/(n_H2 M_H2); eta_HHV = n_H2 HHV/P
- **Assumes:** water electrolysis, 2 electrons per H2; heating values at 25 degC
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_electrolyser', I=192970.66424, N=1, V_cell=2.0)
# -> H2_kg_s = 0.00201588 kg/s, H2_kg_h = 7.25717, O2_kg_s = 0.0159994 kg/s, H2O_kg_s = 0.0180153
kg/s, specific_energy_kWh_per_kg = 53.1807, efficiency_HHV = 0.740594, efficiency_LHV = 0.626587
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Electrolyser: I = 192970.66424 A, N = 1, V_cell = 2 V	SOLVED	H2_kg_s = 0.00201588 kg/s, H2_kg_h = 7.25717, O2_kg_s = 0.0159994 kg/s, H2O_kg_s = 0.0180153 kg/s, specific_energy_kWh_per_kg = 53.1807, efficiency_HHV = 0.740594, efficiency_LHV = 0.626587
Electrolyser: I = 1000 A, N = 100, V_cell = 1 V	INVALID_INPUT	Water electrolyser: hydrogen production and energy per kilogram: an electrolysis cell operates between the reversible vo

Hydrogen stored in a tank at pressure (real gas) — `energy_hydrogen_tank`

Density, compressibility factor, stored mass and energy (lower heating value) of compressed hydrogen in a tank at a pressure and temperature, by the Leachman (2009) equation of state for normal hydrogen, compared with the ideal-gas estimate.

$$p = \rho R T \left(1 + \delta \frac{\partial \alpha^r}{\partial \delta}\right), \quad Z = \frac{p}{\rho R T}$$

input	meaning	condition
`V`	internal tank volume	
`p`	gas pressure (absolute)	
`T`	gas temperature	

output	meaning
`rho`	hydrogen density (real gas)
`Z`	compressibility factor
`mass`	stored hydrogen mass
`rho_ideal`	ideal-gas density p M/(R T)
`mass_ideal`	ideal-gas mass
`ideal_over_real`	ideal-gas over real density (= Z)
`energy_LHV_kWh`	stored chemical energy (LHV) in kWh

- **Method:** Leachman 2009 Helmholtz equation: $p = \rho$ R T (1 + delta d(alpha_r)/d(delta)), solved for rho; Z = p/(rho R T)
- **Assumes:** 1 kPa <= p <= 1500 bar; 200 $K \leq T \leq 700$ K; normal hydrogen, uniform state
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_hydrogen_tank', V=1.0, p=70000000.0, T=288.15)
# -> rho = 40.1722 kg/m^3, Z = 1.46617, mass = 40.1722 kg, rho_ideal = 58.8992 kg/m^3, mass_ideal =
58.8992 kg, ideal_over_real = 1.46617, energy_LHV_kWh = 1338.63   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Hydrogen tank: V = 1 m^3, p = 700 bar, T = 288.15 K	SOLVED	rho = 40.1722 kg/m^3, Z = 1.46617, mass = 40.1722 kg, rho_ideal = 58.8992 kg/m^3, mass_ideal = 58.8992 kg, ideal_over_real = 1.46617, energy_LHV_kWh = 1338.63
Hydrogen tank: V = 1 m^3, p = 700 bar, T = 50 K	INVALID_INPUT	Hydrogen stored in a tank at pressure (real gas): the temperature must lie between 200 K and 700 K (gaseous hydrogen).

Minimum (isothermal) work to compress hydrogen — `energy_hydrogen_isothermal_compression`

The reversible isothermal work per kilogram to compress hydrogen from p1 to p2 at temperature T (the integral of v dp, equal to the Gibbs-energy rise) for the real gas (Leachman equation of state) and the ideal gas.

$$w = \int_{p_1}^{p_2} v\,dp = g(p_2, T) - g(p_1, T), \quad w_{ideal} = \frac{RT}{M}\ln\frac{p_2}{p_1}$$

input	meaning	condition
`p1`	initial pressure	
`p2`	final pressure	
`T`	compression temperature	

output	meaning
`w`	real-gas isothermal work per kg
`w_kWh_per_kg`	the same in kWh/kg
`w_ideal`	ideal-gas work R T ln(p2/p1)/M
`real_over_ideal`	real over ideal work

- **Method:** $w = g$(p2, T) - g(p1, T), $g/(RT) = ln$(delta) + alpha_r + delta d(alpha_r)/d(delta) + (terms in T only)
- **Assumes:** reversible and isothermal; 200 $K \le T \le 700$ K, pressures up to 1500 bar
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_hydrogen_isothermal_compression', p1=100000.0, p2=200000.0,
T=300)
# -> w = 858386 J/kg, w_kWh_per_kg = 0.238441, w_ideal = 857663 J/kg, real_over_ideal = 1.00084
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Hydrogen isothermal compression: p1 = 1 bar, p2 = 2 bar, T = 300 K	SOLVED	w = 858386 J/kg, w_kWh_per_kg = 0.238441, w_ideal = 857663 J/kg, real_over_ideal = 1.00084
Hydrogen isothermal compression: p1 = 700 bar, p2 = 20 bar, T = 300 K	INVALID_INPUT	Minimum (isothermal) work to compress hydrogen: the final pressure must exceed the initial pressure.

Isentropic (adiabatic) compression of hydrogen — `energy_hydrogen_isentropic_compression`

Outlet temperature and work per kilogram of a reversible adiabatic compression of hydrogen from (p1, T1) to p2, from the full Leachman equation of state (real gas, variable heat capacity).

$$s(p_2, T_2) = s(p_1, T_1), \quad w = h_2 - h_1$$

input	meaning	condition
`p1`	inlet pressure	
`T1`	inlet temperature	
`p2`	outlet pressure	

output	meaning
`T2`	outlet temperature
`w`	compression work per kg $(h_2 - h_1)$
`w_kWh_per_kg`	work in kWh/kg

- **Method:** s(p2, T2) = s(p1, T1) solved for T2 by bisection; $w = h$(p2, T2) - h(p1, T1); s and h from the Helmholtz energy (ideal + residual parts)
- **Assumes:** reversible adiabatic, single stage; outlet temperature below 1000 K
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_hydrogen_isentropic_compression', p1=3000000.0, T1=300.0,
p2=10000000.0)
# -> T2 = 423.294 K, w = 1.83321e+06 J/kg, w_kWh_per_kg = 0.509226   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Hydrogen isentropic compression: p1 = 30 bar, T1 = 300 K, p2 = 100 bar	SOLVED	T2 = 423.294 K, w = 1.83321e+06 J/kg, w_kWh_per_kg = 0.509226
Hydrogen isentropic compression: p1 = 30 bar, T1 = 300 K, p2 = 10 bar	INVALID_INPUT	Isentropic (adiabatic) compression of hydrogen: the final pressure must exceed the initial pressure.

PV single-diode model: current at a voltage — `energy_single_diode_current`

The current (and power) of a PV cell or module at a terminal voltage from the five-parameter single-diode equation $I = IL - I_0$ [exp((V + I Rs)/a) - 1] - (V + I Rs)/Rsh, solved implicitly.

$$I = I_L - I_0 \left[e^{(V + I R_s)/a} - 1 \right] - \frac{V + I R_s}{R_{sh}}$$

input	meaning	condition
`V`	terminal voltage	
`IL`	photocurrent (light-generated current)	
`I0`	diode saturation current	

input	meaning	condition
`Rs`	series resistance	
`Rsh`	shunt resistance	
`nNsVth`	modified ideality factor n Ns k T/q	

output	meaning
`I`	terminal current
`P`	power V I

- **Method:** the implicit single-diode equation solved for I by bisection to full precision
- **Assumes:** IL, I0, Rsh, $a > 0$; $Rs \geq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_single_diode_current', V=0.0, IL=5.0, I0=1e-10, Rs=0.0,
Rsh=1000000000.0, nNsVth=1.0)
# -> I = 5 A, P = 0 W   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_single_diode_current', V=1.0, IL=5.0, I0=1e-10, Rs=0.0,
Rsh=100.0, nNsVth=1.0)
# -> I = 4.99 A, P = 4.99 W   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Single-diode current: V = 1 V, IL = 5 A, I0 = 1e-10 A, Rs = 0 ohm, Rsh = 100 ohm, nNsVth = 1 V	SOLVED	I = 4.99 A, P = 4.99 W
Single-diode current: V = 1 V, IL = 5 A, I0 = 1e-10 A, Rs = 0 ohm, Rsh = 0 ohm, nNsVth = 1 V	INVALID_INPUT	PV single-diode model: current at a voltage: the photocurrent, saturation current, shunt resistance and nNsVth must be a

PV single-diode model: voltage at a current — `energy_single_diode_voltage`

The terminal voltage (and power) of a PV cell or module carrying a current, from the five-parameter single-diode equation solved implicitly for V; $I = 0$ gives the open-circuit voltage.

$$I = I_L - I_0 \left[e^{(V + I R_s)/a} - 1 \right] - \frac{V + I R_s}{R_{sh}}$$

input	meaning	condition
`I`	terminal current	
`IL`	photocurrent (light-generated current)	
`I0`	diode saturation current	
`Rs`	series resistance	
`Rsh`	shunt resistance	
`nNsVth`	modified ideality factor n Ns k T/q	

output	meaning
`V`	terminal voltage
`P`	power V I

- **Method:** the implicit single-diode equation solved for V by bisection to full precision
- **Assumes:** IL, I0, Rsh, $a > 0$; $Rs \geq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_single_diode_voltage', I=8.0, IL=9.0, I0=1e-10, Rs=0.3,
Rsh=400.0, nNsVth=1.9)
# -> V = 41.1302 V, P = 329.042 W   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Single-diode voltage: I = 0 A, IL = 5 A, I0 = 1e-10 A, Rs = 0 ohm, Rsh = 1e15 ohm, nNsVth = 1 V	SOLVED	V = 24.6353 V, P = 0 W
Single-diode voltage: I = 0 A, IL = 5 A, I0 = 0 A, Rs = 0 ohm, Rsh = 100 ohm, nNsVth = 1 V	INVALID_INPUT	PV single-diode model: voltage at a current: the photocurrent, saturation current, shunt resistance and nNsVth must be a

PV single-diode model: short-circuit, open-circuit and maximum power point — `energy_single_diode_mpp`

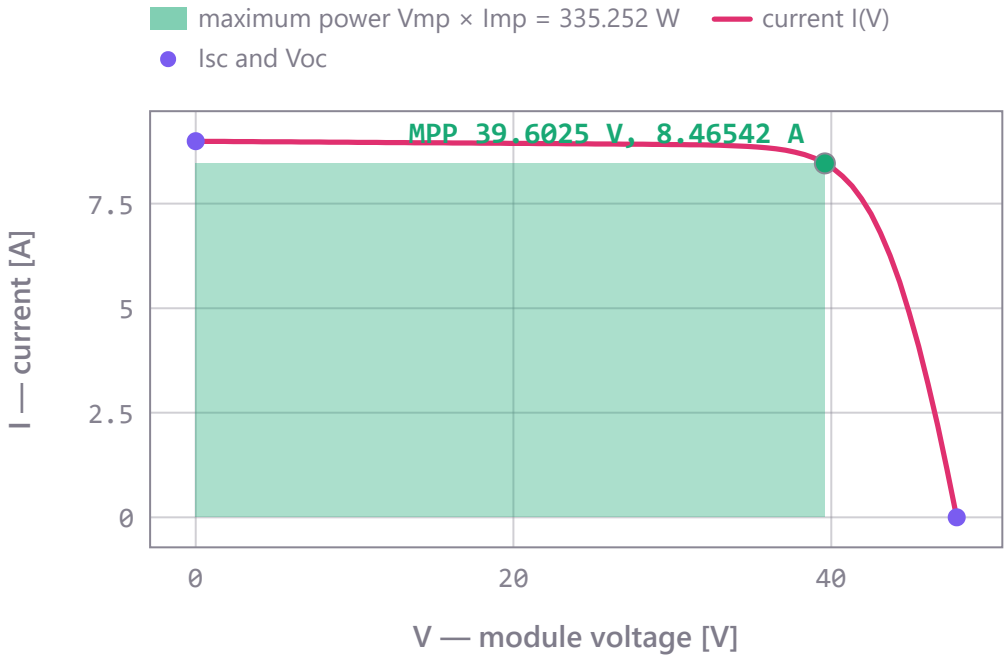
Short-circuit current, open-circuit voltage, the maximum power point (current, voltage, power) and the fill factor of a PV cell or module described by the five-parameter single-diode model.

$$\frac{dP}{dV} = 0, \quad FF = \frac{I_{mp} V_{mp}}{I_{sc} V_{oc}}$$

input	meaning	condition
`IL`	photocurrent (light-generated current)	
`I0`	diode saturation current	
`Rs`	series resistance	
`Rsh`	shunt resistance	
`nNsVth`	modified ideality factor n Ns k T/q	

output	meaning
`Isc`	short-circuit current
`Voc`	open-circuit voltage
`Imp`	current at maximum power
`Vmp`	voltage at maximum power
`Pmp`	maximum power
`FF`	fill factor Pmp/(Isc Voc)

- **Method:** I(0) and V(I = 0) from the implicit equation; maximum power where $dP/dV_d = 0$ along the curve parametrized by the diode voltage (bisection)
- **Assumes:** IL, I0, Rsh, $a > 0$; $Rs \geq 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Current–voltage curve of a PV module (single-diode model: photocurrent 9 A, saturation current 1.0000e-10 A, Rs = 0.3 Ω, Rsh = 400 Ω), computed by the package at 81 voltages. The shaded rectangle is the maximum power the package finds, P = 335.252 W at V = 39.6025 V, I = 8.46542 A; it fills 0.778276 of Isc × Voc (8.99326 A × 47.8984 V), the fill factor.

```
energy_systems_ai999x.solve('energy_single_diode_mpp', IL=9.0, I0=1e-10, Rs=0.3, Rsh=400.0,
nNsVth=1.9)
# -> Isc = 8.99326 A, Voc = 47.8984 V, Imp = 8.46542 A, Vmp = 39.6025 V, Pmp = 335.252 W, FF =
0.778276   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
PV MPP: IL = 5 A, I0 = 1e-10 A, Rs = 0 ohm, Rsh = 1e15 ohm, nNsVth = 1 V	SOLVED	Isc = 5 A, Voc = 24.6353 V, Imp = 4.77798 A, Vmp = 21.5208 V, Pmp = 102.826 W, FF = 0.834788
PV MPP: IL = 5 A, I0 = 1e-10 A, Rs = 0 ohm, nNsVth = 1 V	NEEDS_INPUT	energy_single_diode_mpp needs Rsh (shunt resistance). Example: solve('energy_single_diode_mpp', IL='9 A', I0='1e-10 A',

PV module single-diode parameters at operating conditions (De Soto) — `energy_desoto_parameters`

The five single-diode parameters (photocurrent, saturation current, series and shunt resistance, modified ideality factor) of a PV module at an irradiance and cell temperature from its reference-condition values, by the De Soto (2006) model.

$$I_0 = I_{0,ref}\left(\frac{T}{T_{ref}}\right)^3 \exp\left(\frac{E_{g,ref}}{kT_{ref}} - \frac{E_g}{kT}\right)$$

input	meaning	condition
`G`	effective irradiance	
`T_cell`	cell temperature	
`alpha_sc`	short-circuit current temperature coefficient in A/K	
`a_ref`	modified ideality factor at reference conditions	
`IL_ref`	photocurrent at reference conditions	
`I0_ref`	saturation current at reference conditions	
`Rsh_ref`	shunt resistance at reference conditions	
`Rs`	series resistance	
`EgRef`	band gap at 25 degC in eV (default 1.121, silicon)	
`dEgdT`	band-gap temperature coefficient in $1/K$ (default -0.0002677)	

output	meaning
`IL`	photocurrent
`I0`	saturation current
`Rs`	series resistance
`Rsh`	shunt resistance
`nNsVth`	modified ideality factor

- **Method:** a = a_ref T/T_ref; $IL = (G/1000)$(IL_ref + alpha_sc (T - T_ref)); I0 = I0_ref (T/T_ref)^3 exp(Eg_ref/(k T_ref) - Eg/(k T)), Eg = Eg_ref (1 + dEg/dT (T - T_ref)); Rsh = Rsh_ref $1000/G$
- **Assumes:** reference conditions 1000 W/m^2 and 25 degC
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_desoto_parameters', G=1000, T_cell=298.15, alpha_sc=0.004,
a_ref=1.9, IL_ref=9.0, I0_ref=1e-10, Rsh_ref=400.0, Rs=0.3)
# -> IL = 9 A, I0 = 1e-10 A, Rs = 0.3 ohm, Rsh = 400 ohm, nNsVth = 1.9 V   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
De Soto parameters: G = 1000 W/m^2, T_cell = 25 degC, alpha_sc = 0.004, a_ref = 1.9 V, IL_ref = 9 A, I0_ref = 1e-10 A, Rsh_ref = 400 ohm, Rs = 0.3 ohm	SOLVED	IL = 9 A, I0 = 1e-10 A, Rs = 0.3 ohm, Rsh = 400 ohm, nNsVth = 1.9 V
De Soto parameters: G = 0 W/m^2, T_cell = 25 degC, alpha_sc = 0.004, a_ref = 1.9 V, IL_ref = 9 A, I0_ref = 1e-10 A, Rsh_ref = 400 ohm, Rs = 0.3 ohm	INVALID_INPUT	PV module single-diode parameters at operating conditions (De Soto): the effective irradiance must be a positive number.

PV module maximum power at an irradiance and temperature —

`energy_pv_module_at_conditions`

Short-circuit current, open-circuit voltage, maximum-power current, voltage and power, and fill factor of a PV module at an irradiance and cell temperature: the De Soto translation of its reference parameters, then the single-diode maximum power point.

$$P_{mp} = \max_{V} V\, I(V;\, I_L(G,T), I_0(T), R_s, R_{sh}(G), a(T))$$

input	meaning	condition
`G`	effective irradiance	
`T_cell`	cell temperature	
`alpha_sc`	short-circuit current temperature coefficient in A/K	
`a_ref`	modified ideality factor at reference conditions	
`IL_ref`	photocurrent at reference conditions	
`I0_ref`	saturation current at reference conditions	
`Rsh_ref`	shunt resistance at reference conditions	
`Rs`	series resistance	
`EgRef`	band gap at 25 degC in eV (default 1.121, silicon)	
`dEgdT`	band-gap temperature coefficient in $1/K$ (default -0.0002677)	

output	meaning
`Isc`	short-circuit current
`Voc`	open-circuit voltage
`Imp`	current at maximum power
`Vmp`	voltage at maximum power
`Pmp`	maximum power
`FF`	fill factor

- **Method:** De Soto parameter translation, then I_sc, V_oc and the maximum power point of the single-diode equation
- **Assumes:** reference conditions 1000 W/m^2 and 25 degC
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_pv_module_at_conditions', G=800, T_cell=318.15, alpha_sc=0.004,
a_ref=1.9, IL_ref=9.0, I0_ref=1e-10, Rsh_ref=400.0, Rs=0.3)
# -> Isc = 7.25964 A, Voc = 44.2796 V, Imp = 6.79029 A, Vmp = 36.3735 V, Pmp = 246.987 W, FF =
0.768342   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
PV module at conditions: G = 1000 W/m^2, T_cell = 25 degC, alpha_sc = 0.004, a_ref = 1 V, IL_ref = 5 A, I0_ref = 1e-10 A, Rsh_ref = 1e15 ohm, Rs = 0 ohm	SOLVED	Isc = 5 A, Voc = 24.6353 V, Imp = 4.77798 A, Vmp = 21.5208 V, Pmp = 102.826 W, FF = 0.834788
PV module at conditions: G = 1000 W/m^2, T_cell = 500 K, alpha_sc = 0.004, a_ref = 1 V, IL_ref = 5 A, I0_ref = 1e-10 A, Rsh_ref = 1e15 ohm, Rs = 0 ohm	INVALID_INPUT	PV module maximum power at an irradiance and temperature: the cell temperature must lie between 200 K and 400 K.

Irradiance on a tilted plane (Perez sky model) — `energy_poa_perez`

Beam, sky-diffuse and ground-reflected irradiance on a tilted surface with the Perez (1990) anisotropic sky (circumsolar and horizon brightening coefficients by sky-clearness bin, all-sites composite set), the air mass by Kasten-Young and G_on from day n (Spencer).

$$G_{d,t} = G_d \left[(1 - F_1) \frac{1 + \cos\beta}{2} + F_1 \frac{a}{b} + F_2 \sin\beta \right]$$

input	meaning	condition
`tilt`	surface tilt from horizontal in degrees	
`surface_azimuth`	surface azimuth in degrees clockwise from north (south = 180)	
`zenith`	solar zenith angle in degrees	
`azimuth`	solar azimuth in degrees clockwise from north	
`dni`	direct normal irradiance	
`ghi`	global horizontal irradiance	
`dhi`	diffuse horizontal irradiance	
`albedo`	ground reflectance (0 to 1)	
`n`	day of the year (1 = 1 January)	

output	meaning
`poa_global`	total irradiance on the plane
`poa_direct`	beam irradiance on the plane
`poa_sky_diffuse`	sky-diffuse irradiance on the plane
`poa_ground_diffuse`	ground-reflected irradiance on the plane
`aoi_deg`	angle of incidence in degrees

- **Method:** eps = ((DHI + DNI)/DHI + 1.041 z^3)/(1 + 1.041 z^3), Delta = DHI AM/G_on; F1, F2 from the bin's coefficients; sky = DHI [$(1 - F_1)$(1 + cos beta)/2 + F1 $a/b + F_2$ sin beta]
- **Assumes:** 0 <= zenith < 87; DHI > 0; DHI <= GHI
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_poa_perez', tilt=30, surface_azimuth=180, zenith=30,
azimuth=180, dni=800, ghi=792.820323027551, dhi=100, albedo=0.2, n=172)
# -> poa_global = 923.778 W/m^2, poa_direct = 800 W/m^2, poa_sky_diffuse = 113.156 W/m^2,
poa_ground_diffuse = 10.6218 W/m^2, aoi_deg = 0   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Plane of array irradiance (Perez): tilt = 30, surface_azimuth = 180, zenith = 30, azimuth = 180, dni = 800 W/m^2, ghi = 792.820323027551 W/m^2, dhi = 100 W/m^2, albedo = 0.2, n = 172	SOLVED	poa_global = 923.778 W/m^2, poa_direct = 800 W/m^2, poa_sky_diffuse = 113.156 W/m^2, poa_ground_diffuse = 10.6218 W/m^2, aoi_deg = 0

question	outcome	result
Plane of array irradiance (Perez): tilt = 30, surface_azimuth = 180, zenith = 30, azimuth = 180, dni = 800 W/m^2, ghi = 792.820323027551 W/m^2, dhi = 0 W/m^2, albedo = 0.2, n = 172	INVALID_INPUT	Irradiance on a tilted plane (Perez sky model): the Perez model needs a positive diffuse irradiance.

Clear-sky irradiance (Ineichen-Perez with Linke turbidity) — `energy_clear_sky_ineichen`

Clear-sky global horizontal, direct normal and diffuse horizontal irradiance from the apparent solar zenith, the Linke turbidity and the site altitude (Ineichen and Perez 2002), with the pressure-corrected air mass and G_on from day n.

$$G_{cs} = c_{g1} G_{on} \cos z \, e^{-c_{g2} AM (f_{h1} + f_{h2} (T_L - 1))}$$

input	meaning	condition
`zenith`	apparent solar zenith angle in degrees	
`linke_turbidity`	Linke turbidity factor (1 to 10; about 2 to 5 for clear skies)	
`altitude`	site altitude above sea level	
`n`	day of the year (1 = 1 January)	

output	meaning
`ghi`	clear-sky global horizontal irradiance
`dni`	clear-sky direct normal irradiance
`dhi`	clear-sky diffuse horizontal irradiance
`AM_absolute`	absolute air mass (Kasten-Young, standard pressure at the altitude)

- **Method:** GHI = cg1 G_on cos z exp(-cg2 AM (fh1 + fh2 $(TL - 1)$)); DNI = min(b G_on exp(-0.09 AM $(TL - 1)$), GHI (1 - (0.1 - 0.2 e^-TL)/(0.1 + 0.882/fh1))/cos z); $fh1 = e^{-h/8000}$, $fh2 = e^{-h/1250}$
- **Assumes:** sun above the horizon; $1 \le TL \le 10$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_clear_sky_ineichen', zenith=0, linke_turbidity=1, altitude=0, n=1)
# -> ghi = 1180.76 W/m^2, dni = 1148.99 W/m^2, dhi = 31.7724 W/m^2, AM_absolute = 0.999712 (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Clear-sky irradiance (Ineichen): zenith = 30, linke_turbidity = 3, altitude = 500 m, n = 172	SOLVED	ghi = 885.003 W/m^2, dni = 910.142 W/m^2, dhi = 96.7974 W/m^2, AM_absolute = 1.08721
Clear-sky irradiance (Ineichen): zenith = 95, linke_turbidity = 3, altitude = 500 m, n = 172	INVALID_INPUT	Clear-sky irradiance (Ineichen-Perez with Linke turbidity): the sun must be above the horizon (apparent zenith below 90
Clear-sky irradiance (Ineichen): zenith = 30, linke_turbidity = 3, n = 172	NEEDS_INPUT	energy_clear_sky_ineichen needs altitude (site altitude above sea level). Example: solve('energy_clear_sky_ineichen', ze

Direct normal irradiance from global horizontal irradiance (DISC model) — `energy_disc_dni`

The direct normal irradiance estimated from global horizontal irradiance by Maxwell's DISC model (clearness index and pressure-corrected air mass), with the clearness index and the air mass used.

$$G_{bn} = I_0 \left[K_{nc}(AM) - \left(a + b\, e^{c\, AM} \right) \right]$$

input	meaning	condition
`ghi`	global horizontal irradiance	
`zenith`	solar zenith angle in degrees	
`n`	day of the year (1 = 1 January)	
`pressure`	site pressure (default 101325 Pa)	

output	meaning
`dni`	direct normal irradiance (0 when the model gives a negative value)
`kt`	clearness index (solar constant 1370 W/m^2)
`airmass`	absolute air mass (Kasten 1966, capped at 12)

- **Method:** kt = GHI/(I0 cos z); Kn = Knc(AM) - (a(kt) + b(kt) exp(c(kt) AM)); DNI = Kn I0
- **Assumes:** 0 <= zenith <= 85; GHI > 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_disc_dni', ghi=709.0092500000001, zenith=0, n=1)
# -> dni = 163.645 W/m^2, kt = 0.5, airmass = 0.999494   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
DISC model: ghi = 709.0092500000001 W/m^2, zenith = 0, n = 1	SOLVED	dni = 163.645 W/m^2, kt = 0.5, airmass = 0.999494
DISC model: ghi = 600 W/m^2, zenith = 89, n = 172	INVALID_INPUT	Direct normal irradiance from global horizontal irradiance (DISC model): the zenith angle must lie between 0 and 85 degr

Base impedance and per-unit impedance — `energy_per_unit_impedance`

The base impedance Z_base = V_base^2/S_base and base current of a three-phase per-unit system, and an impedance in ohms expressed in per unit.

$$Z_{base} = \frac{V_{base}^2}{S_{base}}, \quad Z_{pu} = \frac{Z}{Z_{base}}$$

input	meaning	condition
`R`	resistance per phase (ohm)	
`X`	reactance per phase (ohm)	
`S_base`	three-phase base power (VA)	

input	meaning	condition
`V_base`	line-to-line base voltage	

output	meaning
`Z_base`	base impedance V_base^2/S_base
`I_base`	base current S_base/(sqrt(3) V_base)
`R_pu`	per-unit resistance
`X_pu`	per-unit reactance
`Z_pu`	per-unit impedance magnitude

- **Method:** Z_base = V_base^2/S_base, I_base = S_base/(sqrt(3) V_base), Z_pu = Z/Z_base
- **Assumes:** R, $\boldsymbol{X} \geq \mathbf{0}$, not both 0; positive bases; three-phase base, line-to-line voltage
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_per_unit_impedance', R=5.0, X=20.0, S_base=100000000.0,
V_base=132000.0)
# -> Z_base = 174.24 ohm, I_base = 437.387 A, R_pu = 0.0286961, X_pu = 0.114784, Z_pu = 0.118317
(SOLVED)
```

```
energy_systems_ai999x.solve('energy_per_unit_impedance', R=0.0, X=10.0, S_base=1000000.0,
V_base=10000.0)
# -> Z_base = 100 ohm, I_base = 57.735 A, R_pu = 0, X_pu = 0.1, Z_pu = 0.1   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Per-unit impedance: R = 5 ohm, X = 20 ohm, S_base = 100 MVA, V_base = 132 kV	SOLVED	Z_base = 174.24 ohm, I_base = 437.387 A, R_pu = 0.0286961, X_pu = 0.114784, Z_pu = 0.118317
Per-unit impedance: R = 5 ohm, X = 20 ohm, S_base = 0 MVA, V_base = 132 kV	INVALID_INPUT	Base impedance and per-unit impedance: the base power and base voltage must be a positive number.
Per-unit impedance: R = 5 ohm, X = 20 ohm, S_base = 100 MVA	NEEDS_INPUT	energy_per_unit_impedance needs V_base (line-to-line base voltage). Example: solve('energy_per_unit_impedance', R='5 ohm

Geometric mean distance of a three-phase line — `energy_geometric_mean_distance`

The geometric mean distance GMD = (D_ab D_bc D_ca)^(1/3) between the phases of a fully transposed three-phase line.

$$GMD = \sqrt[3]{D_{ab} D_{bc} D_{ca}}$$

input	meaning	condition
`D_ab`	distance between phases a and b	
`D_bc`	distance between phases b and c	
`D_ca`	distance between phases c and a	

output	meaning
`GMD`	geometric mean distance
`ratio_to_mean`	GMD divided by the arithmetic mean spacing

- **Method:** GMD = (D_ab D_bc D_ca)^(1/3)
- **Assumes:** positive spacings that form a triangle (or a straight line); fully transposed line
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_geometric_mean_distance', D_ab=4.0, D_bc=4.0, D_ca=8.0)
# -> GMD = 5.03968 m, ratio_to_mean = 0.944941   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_geometric_mean_distance', D_ab=2.0, D_bc=2.0, D_ca=2.0)
# -> GMD = 2 m, ratio_to_mean = 1   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Geometric mean distance: D_ab = 4 m, D_bc = 4 m, D_ca = 8 m	SOLVED	GMD = 5.03968 m, ratio_to_mean = 0.944941
Geometric mean distance: D_ab = 1 m, D_bc = 1 m, D_ca = 8 m	INVALID_INPUT	Geometric mean distance of a three-phase line: the three spacings must form a triangle (or a line).
Geometric mean distance: D_ab = 4 m, D_bc = 4 m	NEEDS_INPUT	energy_geometric_mean_distance needs D_ca (distance between phases c and a). Example: solve('energy_geometric_mean_dista

Synchronous speed, rotor speed and slip of an AC machine — `energy_synchronous_speed`

The synchronous speed $n_s = 120$ f/P of an AC machine and, with the slip, the rotor speed $n = n_s\,(1 - s)$ (or, with the rotor speed, the slip) and the rotor (slip) frequency.

$$n_s = \frac{120 f}{P}, \quad n = n_s(1 - s)$$

input	meaning	condition
`f`	supply frequency	
`poles`	number of poles (even)	
`s`	slip (fraction or %, optional)	
`n_r`	rotor speed (rpm, optional)	

output	meaning
`n_sync_rpm`	synchronous speed in rpm
`omega_sync`	synchronous speed in rad/s
`n_rotor_rpm`	rotor speed in rpm
`slip`	slip (n_s - n)/n_s
`slip_frequency`	rotor current frequency s f

- **Method:** $n_s = 120$ f/P; $n = n_s\,(1 - s)$; $s = (n_s - n)/n_s$; $f_r = s$ f

- **Assumes:** $f > 0$; even number of poles; at most one of slip and rotor speed
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_synchronous_speed', f=50.0, poles=4)
# -> n_sync_rpm = 1500, omega_sync = 157.08 rad/s   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_synchronous_speed', f=60.0, poles=4, s=0.03)
# -> n_sync_rpm = 1800, omega_sync = 188.496 rad/s, n_rotor_rpm = 1746, slip = 0.03, slip_frequency
= 1.8 Hz   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_synchronous_speed', f=60.0, poles=6, n_r=120.42771838760873)
# -> n_sync_rpm = 1200, omega_sync = 125.664 rad/s, n_rotor_rpm = 1150, slip = 0.0416667,
slip_frequency = 2.5 Hz   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Synchronous speed: f = 50 Hz, poles = 4	SOLVED	n_sync_rpm = 1500, omega_sync = 157.08 rad/s
Synchronous speed: f = 60 Hz, poles = 4, s = 3%	SOLVED	n_sync_rpm = 1800, omega_sync = 188.496 rad/s, n_rotor_rpm = 1746, slip = 0.03, slip_frequency = 1.8 Hz
Synchronous speed: f = 60 Hz, poles = 6, n_r = 1150 rpm	SOLVED	n_sync_rpm = 1200, omega_sync = 125.664 rad/s, n_rotor_rpm = 1150, slip = 0.0416667, slip_frequency = 2.5 Hz
Synchronous speed: f = 50 Hz, poles = 5	INVALID_INPUT	Synchronous speed, rotor speed and slip of an AC machine: the number of poles must be an even whole number from 2 to 200
Synchronous speed: f = 50 Hz	NEEDS_INPUT	energy_synchronous_speed needs poles (number of poles (even)). Example: solve('energy_synchronous_speed', f='50 Hz', pol

Electrical power of a fuel-cell or electrolyser stack — `energy_stack_electrical_power`

The electrical power $P = N$ I V_cell delivered by a fuel-cell stack (or drawn by an electrolyser stack) of N cells in series at a current and an average cell voltage, and the stack voltage.

$$P = N\,I\,V_{cell}$$

input	meaning	condition
`I`	stack current	
`N`	number of cells in series	
`V_cell`	average cell voltage	

output	meaning
`P`	electrical power N I V_cell
`P_kW`	electrical power in kW
`V_stack`	stack voltage N V_cell

- **Method:** $P = N$ I V_cell, V_stack = N V_cell

- **Assumes:** cells in series carrying the same current; 0 < V_cell <= 5 V
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_stack_electrical_power', I=200.0, N=100, V_cell=0.7)
# -> P = 14000 W, P_kW = 14, V_stack = 70 V   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_stack_electrical_power', I=1000.0, N=100, V_cell=1.9)
# -> P = 190000 W, P_kW = 190, V_stack = 190 V   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Stack electrical power: I = 200 A, N = 100, V_cell = 0.7 V	SOLVED	P = 14000 W, P_kW = 14, V_stack = 70 V
Stack electrical power: I = 200 A, N = 0, V_cell = 0.7 V	INVALID_INPUT	N: must be at least 1, got 0.
Stack electrical power: I = 200 A, N = 100	NEEDS_INPUT	energy_stack_electrical_power needs V_cell (average cell voltage). Example: solve('energy_stack_electrical_power', I='20

State of charge by coulomb counting — `energy_coulomb_counting`

The state of charge of a battery after a sequence of constant currents over equal time steps (positive on discharge), by coulomb counting SOC = SOC0 - integral(I dt)/Q, with its lowest and highest value and the charge delivered.

$$SOC(t) = SOC_0 - \frac{1}{Q}\int_0^t I\,dt$$

input	meaning	condition
`SOC0`	initial state of charge (fraction or %)	
`Q`	capacity (Ah, mAh, C)	
`currents`	current in each time step in A (positive on discharge)	
`dt`	length of each time step	

output	meaning
`SOC`	final state of charge
`SOC_min`	lowest state of charge
`SOC_max`	highest state of charge
`charge_out`	net charge delivered
`charge_out_Ah`	net charge delivered in Ah

- **Method:** SOC_k = SOC0 - (sum_{$j \le k$} I_j dt)/Q
- **Assumes:** 0 <= SOC0 <= 1; Q, $dt > 0$; the state of charge stays within 0 to 100%; no coulombic losses
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)

```
energy_systems_ai999x.solve('energy_coulomb_counting', SOC0=0.9, Q=180000.0, currents=[10, 20, -5],
dt=3600.0)
# -> SOC = 0.4, SOC_min = 0.3, SOC_max = 0.9, charge_out = 90000 C, charge_out_Ah = 25   (SOLVED)
```

```
energy_systems_ai999x.solve('energy_coulomb_counting', SOC0=1.0, Q=7200.0, currents='1, 1',
dt=1800.0)
# -> SOC = 0.5, SOC_min = 0.5, SOC_max = 1, charge_out = 3600 C, charge_out_Ah = 1   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Coulomb counting: SOC0 = 90%, Q = 50 Ah, currents = [10, 20, -5], dt = 1 h	SOLVED	SOC = 0.4, SOC_min = 0.3, SOC_max = 0.9, charge_out = 90000 C, charge_out_Ah = 25
Coulomb counting: SOC0 = 90%, Q = 50 Ah, currents = [100, 20], dt = 1 h	INVALID_INPUT	State of charge by coulomb counting: the state of charge leaves the range 0 to 100% (more charge than the capacity).
Coulomb counting: SOC0 = 90%, Q = 50 Ah, currents = [10, 20, -5]	NEEDS_INPUT	energy_coulomb_counting needs dt (length of each time step). Example: solve('energy_coulomb_counting', SOC0=0.9, Q='1800